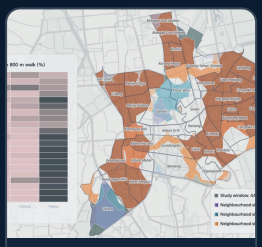
MapX

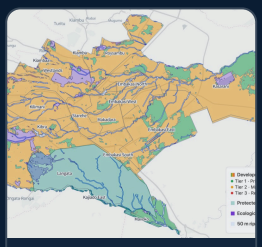
MapX built-in scenarios · urban and rural planning

Scenario Handbook

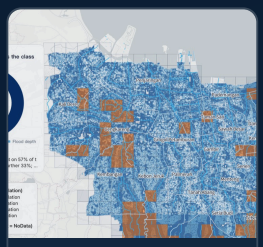
Six planning scenarios: study design, data and methods



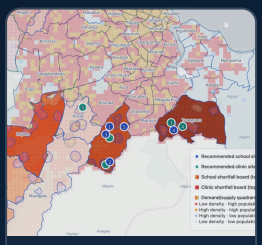
15-minute Neighbourhood Coverage



Development Suitability Baseline



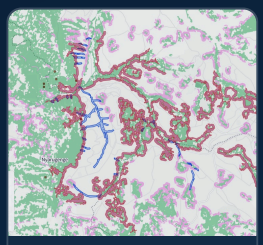
Carrying Capacity Assessment



Public Facility Siting



Urban Growth Monitor



Land-use Conflict Screening

First edition · October 2026

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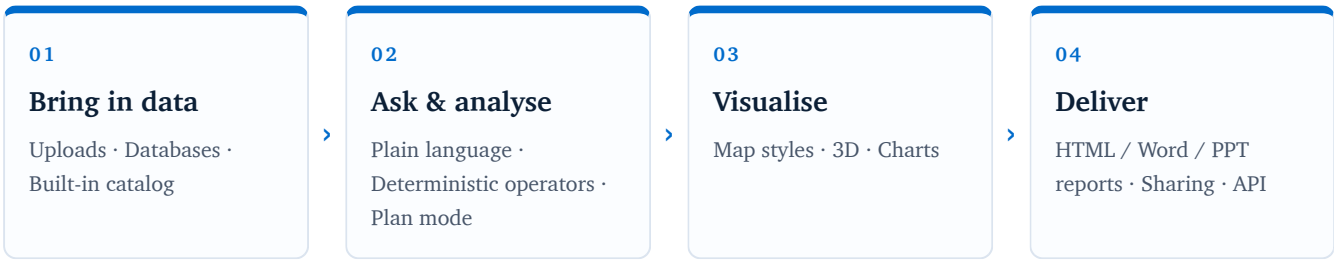
Where development collides with ecological space, steep slopes and riverbanks

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Foreword

MapX is an AI-powered spatial analysis platform that runs in the browser: bring in data, ask questions in plain language, and the platform picks and runs the right spatial operations, rendering results as interactive map layers in seconds — no GIS software to install, no weeks of training.

Data access, spatial analysis, style editing, charts, reports, sharing and collaboration all happen inside one project, and developers can wire the same capabilities into their own products through the REST API, CLI, MCP and skills. The six scenarios in this handbook are the fastest way to experience that workflow.



MapX platform analysis flow

What this handbook covers

This handbook collects the six built-in urban and rural planning scenarios in MapX. Each scenario has one chapter, and every chapter follows the same structure: the planning question, the data it ships, how the study works, how to run it, and what one example run produced.

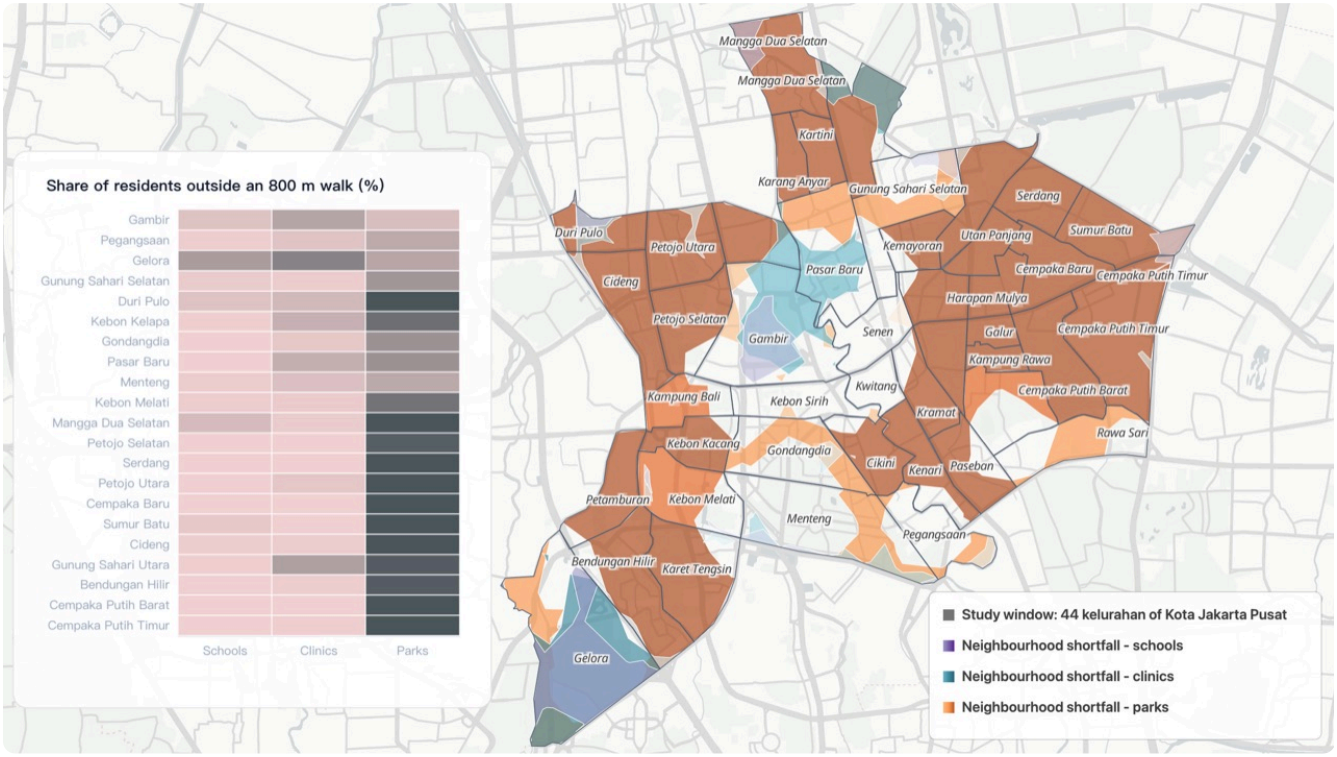
Type	Scenario (study area)	Problem it solves	Deliverables	Time on MapX
Public services	15-minute Neighbourhood Coverage (Jakarta)	Who is covered by schools, clinics and green space within walking distance, and who is not	Three shortfall layers, a neighbourhood table, rankings and a heatmap, a report	~1 hour
Planning assessment	Development Suitability Baseline (Nairobi)	Which land is buildable once ecology, slope and flood constraints are applied	Suitability classes, constraint layers, capacity tables, charts and a report	~1 hour
Planning assessment	Carrying Capacity Assessment (Jakarta)	How much development the water, climate and flood-hazard constraints allow	Capacity classes, overload cells, constraint tables, charts and a report	~1 hour
Public services	Public Facility Siting (Dar es Salaam)	Where coverage gaps are, and which candidate sites close them	Shortfall rankings, candidate cells, recommended sites, a marginal curve and a report	~1 hour
Urban monitoring	Urban Growth Monitor (Hanoi)	How fast, in which direction, and how compactly the built-up area is expanding	Expansion-form layers, phase tables, quadrant grid, charts and a report	~1 hour
Ecological compliance	Land-use Conflict Screening (Kigali)	Where development collides with ecological space, steep slopes and riverbanks	Four conflict layers, a worst-patch ranking, an exposure grid and a report	~1 hour

Solve times are empirical estimates from example runs (plan review, analysis and report generation included); a conventional GIS workflow typically takes person-days to weeks for the same study.

SCENARIO 01

15-minute Neighbourhood Coverage Jakarta

Audit who can reach a school, a clinic and a park within an 800 m walk in central Jakarta, rank the neighbourhoods with the biggest shortfall, and score accessibility.



The Jakarta 15-minute neighbourhood run as delivered: three neighbourhood shortfall ramps, ADM4 boundaries and the pinned three-class comparison heatmap

AT A GLANCE

STUDY AREA **Jakarta**	ENTRY LAYERS **7 layers**
KEY METHODS **Walking service areas · shortfall population · 2SFCA**	DELIVERABLES **Three shortfall layers, a neighbourhood table, rankings and a heatmap, a report**

15-minute Neighbourhood Coverage measures the daily-life side of planning: **who can reach a school, a clinic and a park on foot**, and which neighbourhoods carry the largest shortfall. It is the MapX version of a 15-minute neighbourhood audit, and it deliberately reports two yardsticks — coverage and accessibility — instead of a single score.

What the scenario ships

Seven prepared layers cover Jakarta; the scenario's default study window is Kota Jakarta Pusat (central Jakarta).

Layer	Data	Role in the study
Education facilities	OSM education POIs (7,470 city-wide)	Schools that supply the education circle
Health facilities	Healthsites.io (2,447 city-wide)	Clinics and hospitals that supply the health circle
Road network	Overture transportation	The walking network used for service areas
Administrative boundaries	46 kecamatan (ADM3)	District framing and attribution
Green space and parks	OSM green polygons (3,942)	Source of the destination-park set
Neighbourhood boundaries	kelurahan (ADM4); 44 in the study area, 1.09 km² average	The evaluation unit that is ranked and named
Population	WorldPop, 100 m	Residents at risk of being outside a circle

*All layers come from the **built-in catalog** with source and attribution attached, and they are not limited to Jakarta: the catalog provides the same datasets for other cities at the same specification — switch the study area, add the same entries, and this chapter's method chain transfers in minutes.*

How the study works

Stage	Question	What MapX runs	Output
1. Coverage audit	How much of each neighbourhood is outside the walk?	Select features, Overlay clip/dissolve/erase, Network service area at 800 m, Zonal statistics, Field calculator	Shortfall layer per class with `gap_sum` and a share-of-residents column
2. Service mix and balance	Which neighbourhoods lack one, two or all three classes?	Zonal summary chaining, Overlay dissolve/erase	Neighbourhood summary table with population, coverage count 0–3 and three shortfall columns
3. Accessibility diagnosis	Where is access hardest, even inside the circle?	2SFCA accessibility, Top N ranking, Label anchors, Charts	Top-10 shortfall rankings, per-class accessibility scores and a pinned heatmap

In short, the study first computes a walking service area for each of the three facility classes and crosses it with neighbourhood population to produce three shortfall layers of residents outside the circle; it then assembles the shortfalls into one neighbourhood table and ranks the neighbourhoods with the biggest gaps; finally a 2SFCA accessibility score measures how tight supply is inside the circle, cross-checking coverage against accessibility, and the run delivers rankings, a heatmap and a report.

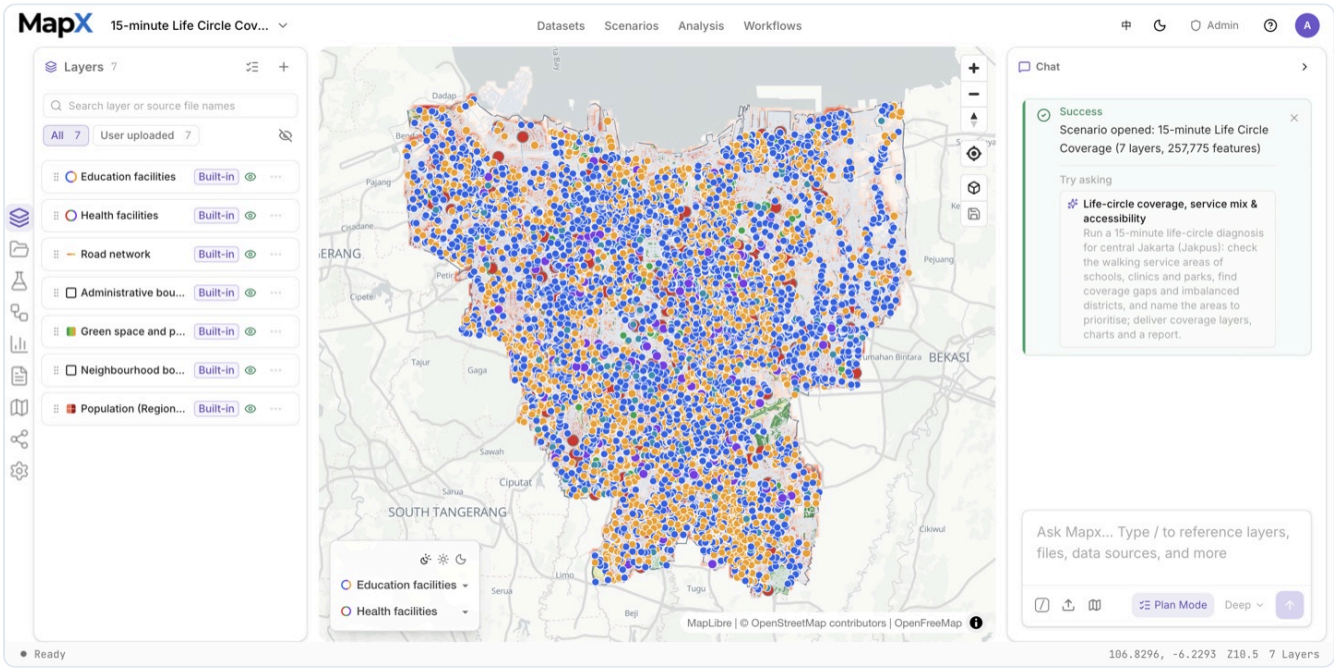
Run it

The screenshots below come from this scenario's reference workspace. Each step names the platform surface it uses, so you can follow the same path in your own project.

> **SUGGESTED QUESTION**
>
> *"Run a 15-minute neighbourhood diagnosis for central Jakarta (Jakpus): check the walking service areas of schools, clinics and parks, find coverage gaps and imbalanced districts, and name the areas to prioritise; deliver coverage layers, charts and a report."*

1. Open the scenario

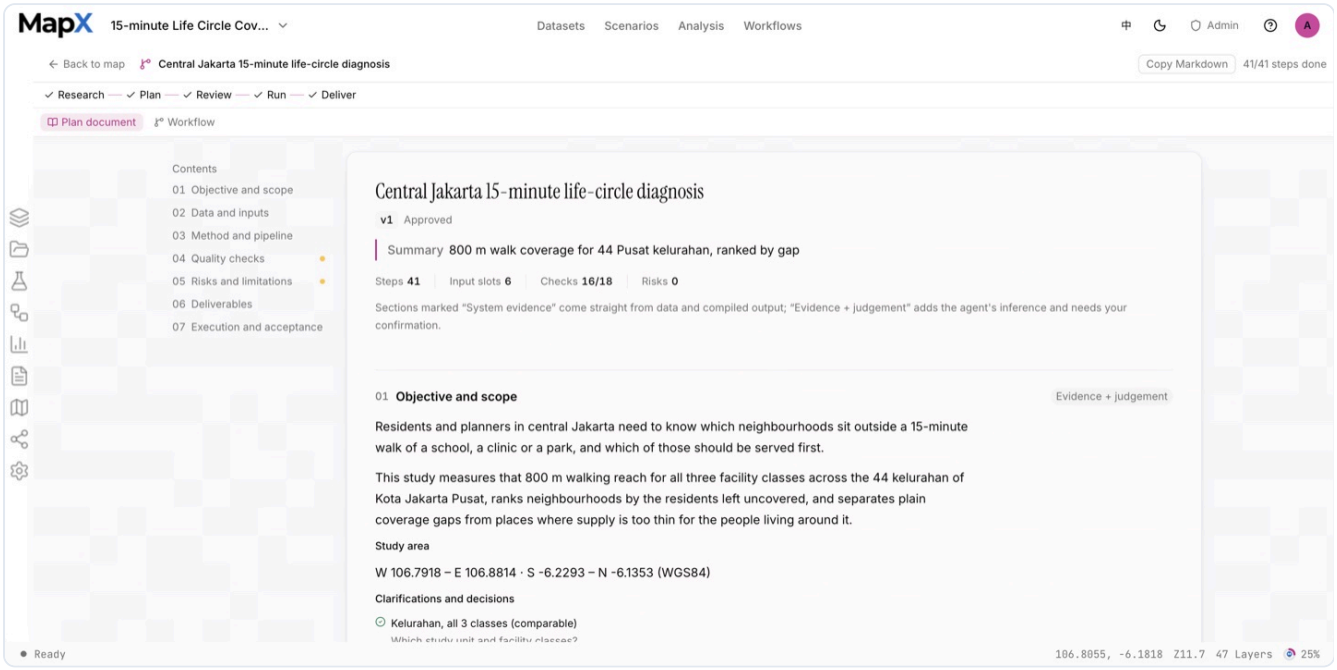
Open **Scenarios** in the top menu, select this card and choose **Open scenario**. The whole bundle imports at once: every layer keeps its default style, legend, source and attribution, and the camera flies to the study area. Use the **Layers** panel to switch layers on and off before you type anything.



The Jakarta scenario opened in a fresh project: the layer bundle imported, the Layers panel expanded, and the chat panel confirming the import with the suggested question

2. Ask the suggested question and review the plan

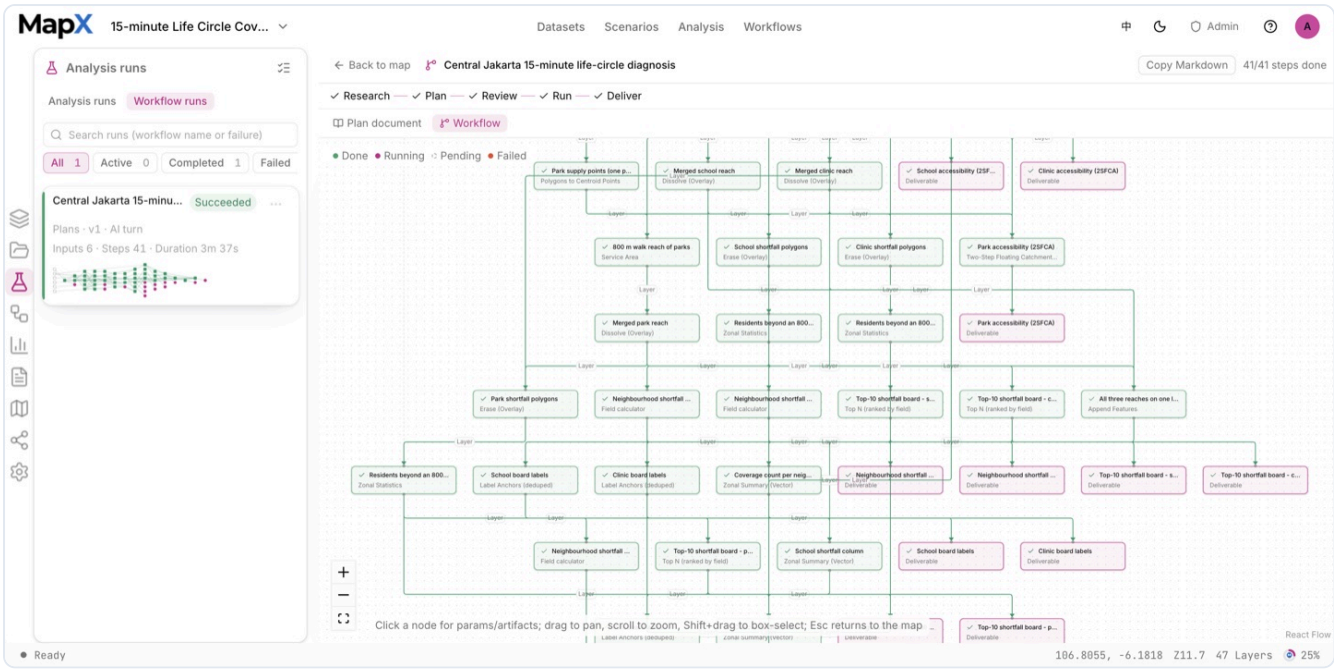
Send the **suggested question** in the box above — or edit it first. **Plan mode** asks the four decisions that change the answer — study unit, walking radius, supply assumption and deliverables — and fixes the rest, including the park definition, in the plan document.



The Jakarta plan document in full-width view: side panels collapsed, plan sections and recorded decisions readable

3. Approve the plan and run the analysis

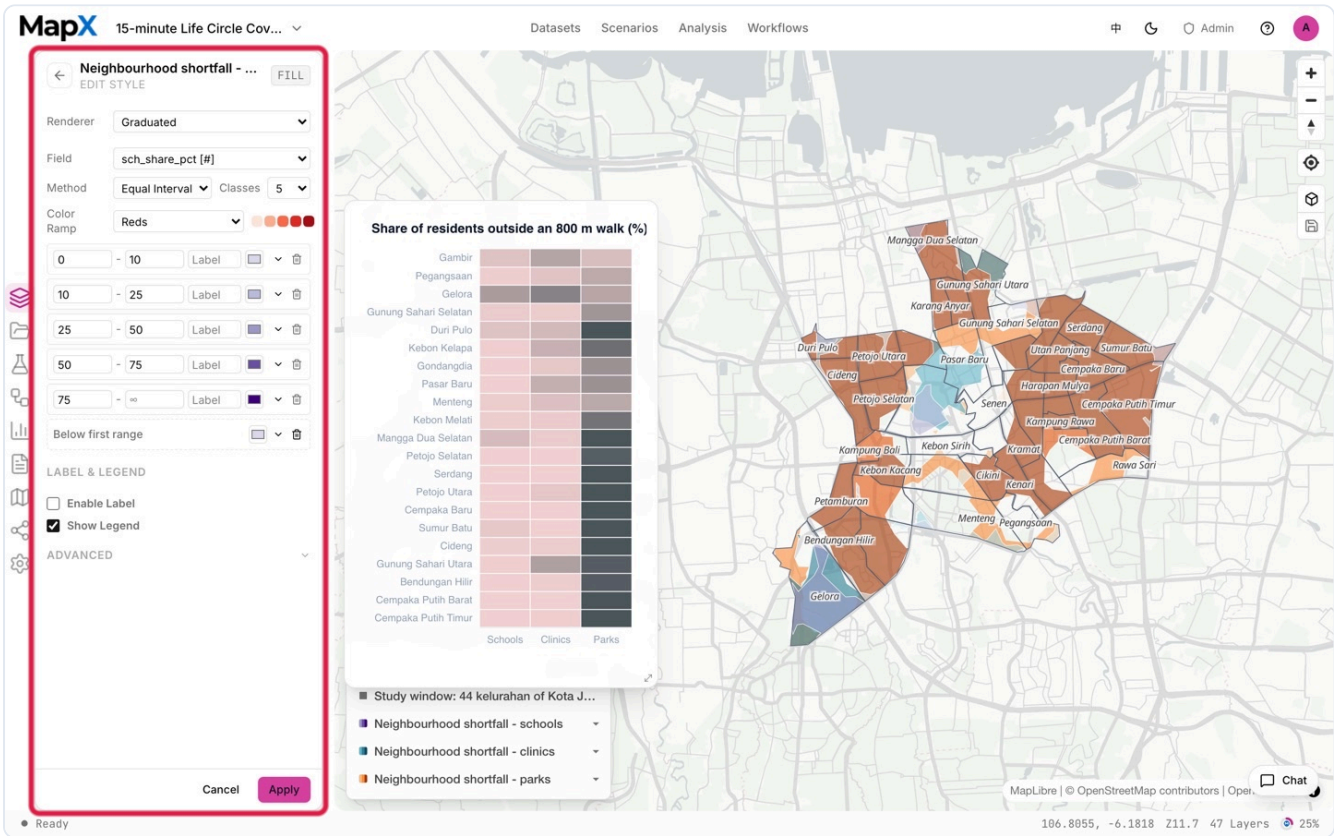
The run uses selection, clipping and erasing to shape the study area, network **service areas** at 800 m for all three facility classes, zonal statistics to count the residents left outside, field calculations for the shortfall shares, the accessibility model (2SFCA) and a chained summary table. The graph shows every branch, including the three facility classes side by side. The **Analysis** panel keeps every run's operator, inputs, parameters and result layer.



The analysis view for Jakarta: the Analysis panel lists every operator run on the left, with the workflow graph in the centre

4. Style the result on the map

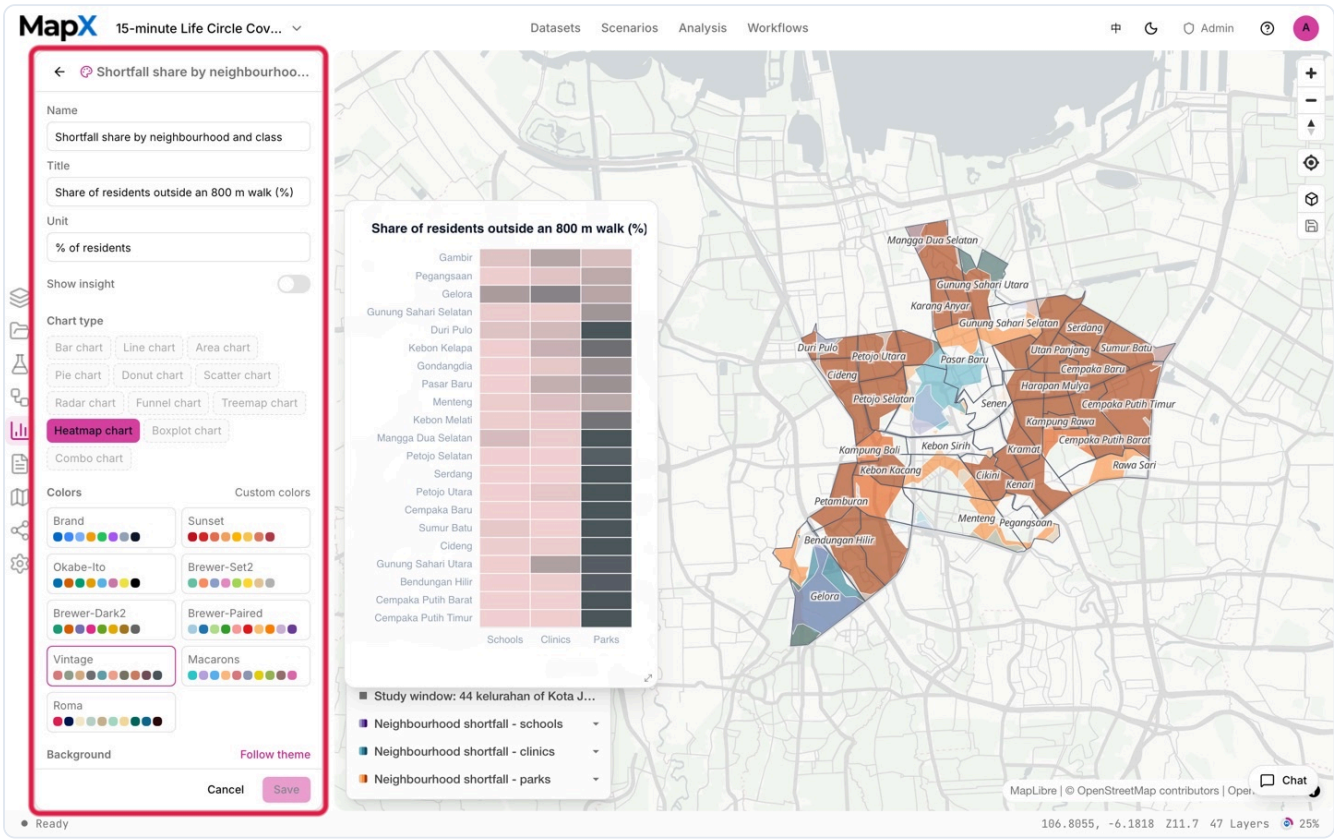
The three shortfall ramps share one break scale, so a dark purple and a dark teal mean the same share; the ADM4 boundary sits on top with neighbourhood labels, and blank space means inside the circle.



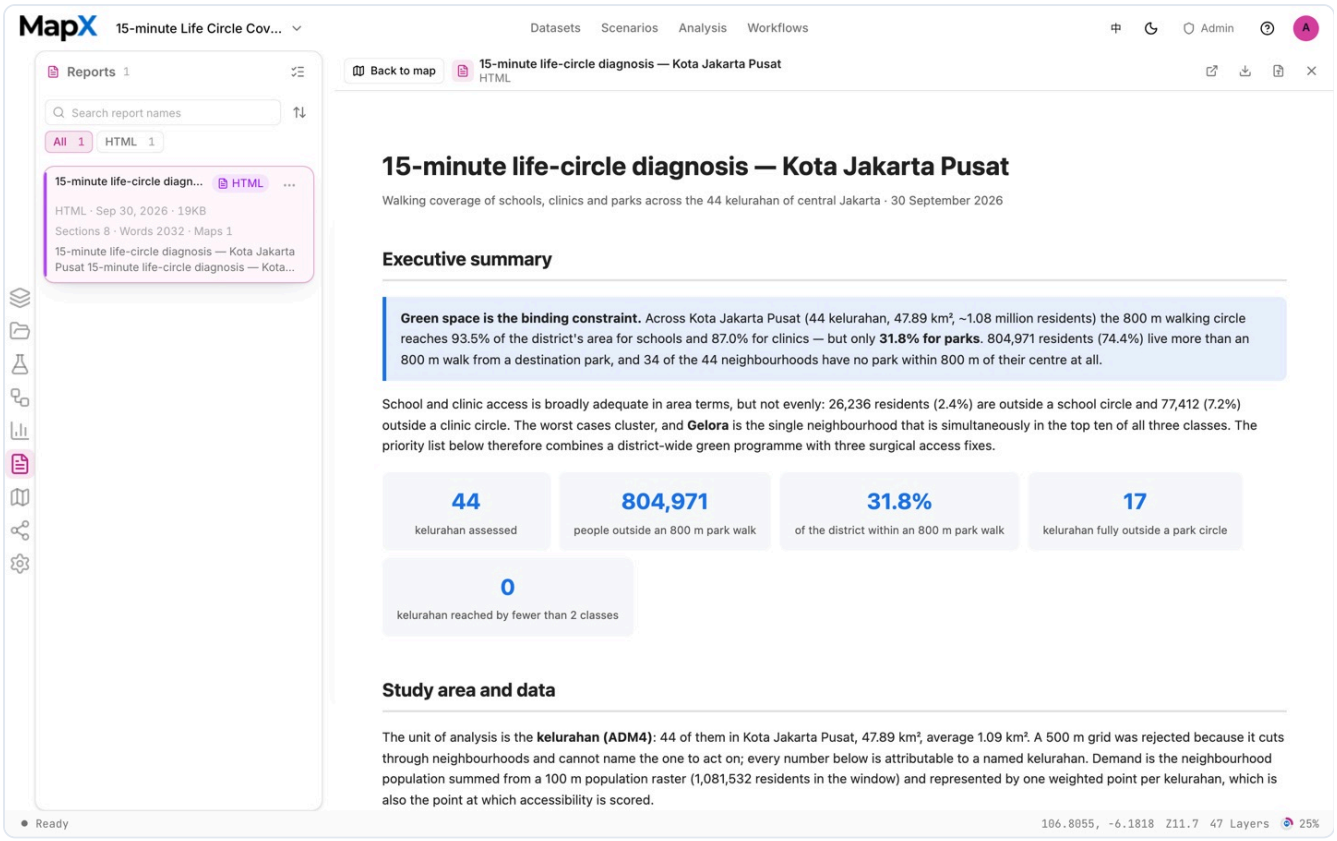
The style editor on a deliverable layer: classed-colour breaks and label settings, with the layer editing panel highlighted in red

5. Turn the result into charts and a report

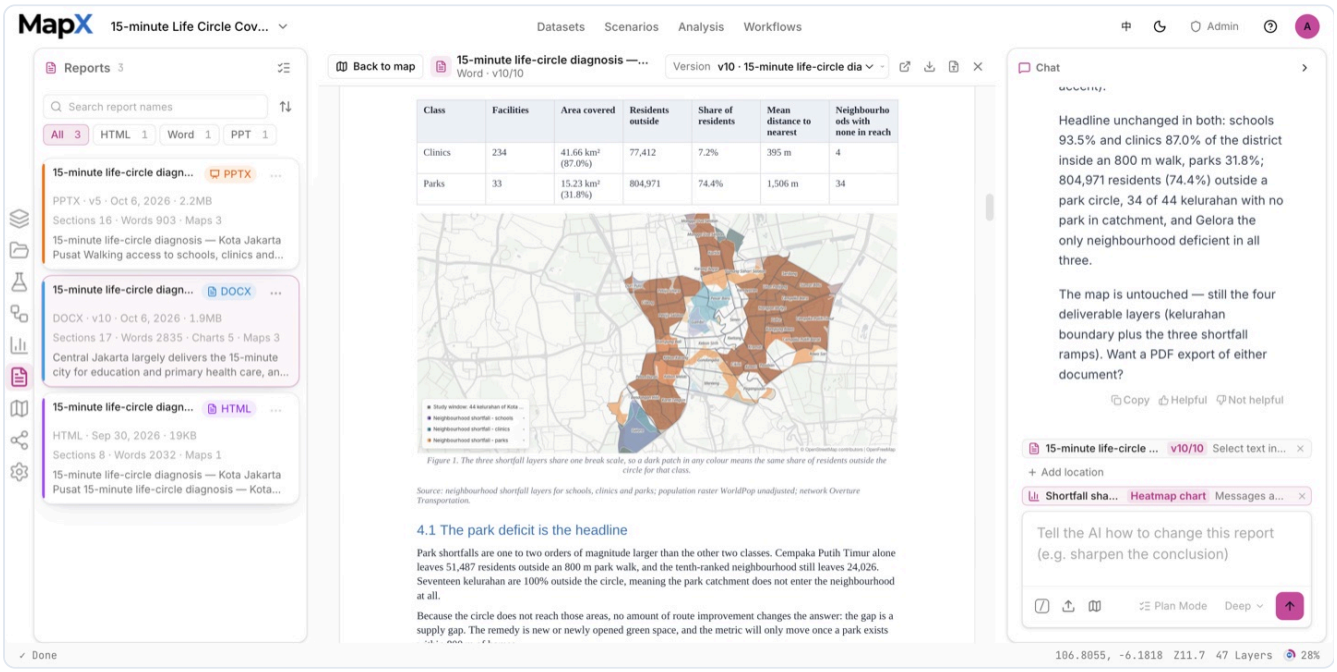
The run produces per-class rankings, a three-class comparison and a heatmap that can be pinned to the map while you pan the ramps. The **report** then names the priority neighbourhoods and states the radius, the park definition and the caveats. The same run can also export the report as a Word document for a formal submission or a slide deck for a presentation handoff.



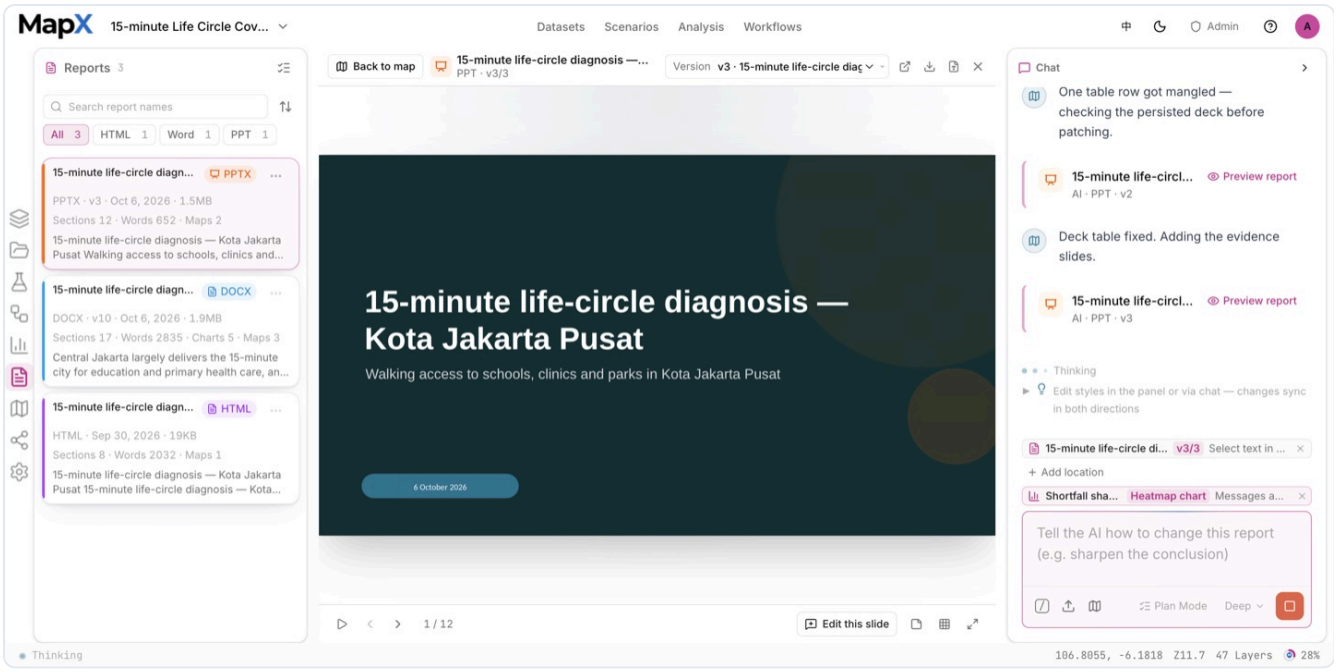
A chart from this scenario in the chart editor, with the editing area highlighted in red



The scenario report opened in the report canvas, with an executive summary, figures and embedded maps



The Word version of the same report: the document preview with the summary table, an embedded map figure and the written analysis



The slide-deck report generated from this run: the title slide in the presentation preview, with the report versions listed in the left panel

Example output

> **NOTE**
>
> The screenshots above come from one example run with the default settings on the shipped data. They show the shape of the deliverables, not what your run will return.

A finished run hands back a named, reusable set of deliverables instead of a single map:

- **Layers** — one neighbourhood shortfall layer per facility class, the accessibility scores, the top-10 rankings, the neighbourhood summary table as a layer, and the labelled boundary.
- **Tables** — the neighbourhood summary with population, coverage count and the three shortfall columns, plus the per-class rankings.
- **Charts** — three class rankings, a grouped comparison and a heatmap, all editable and pinnable.
- **Report** — the diagnosis document naming the priority neighbourhoods, with the radius, the park definition and the coverage caveats.

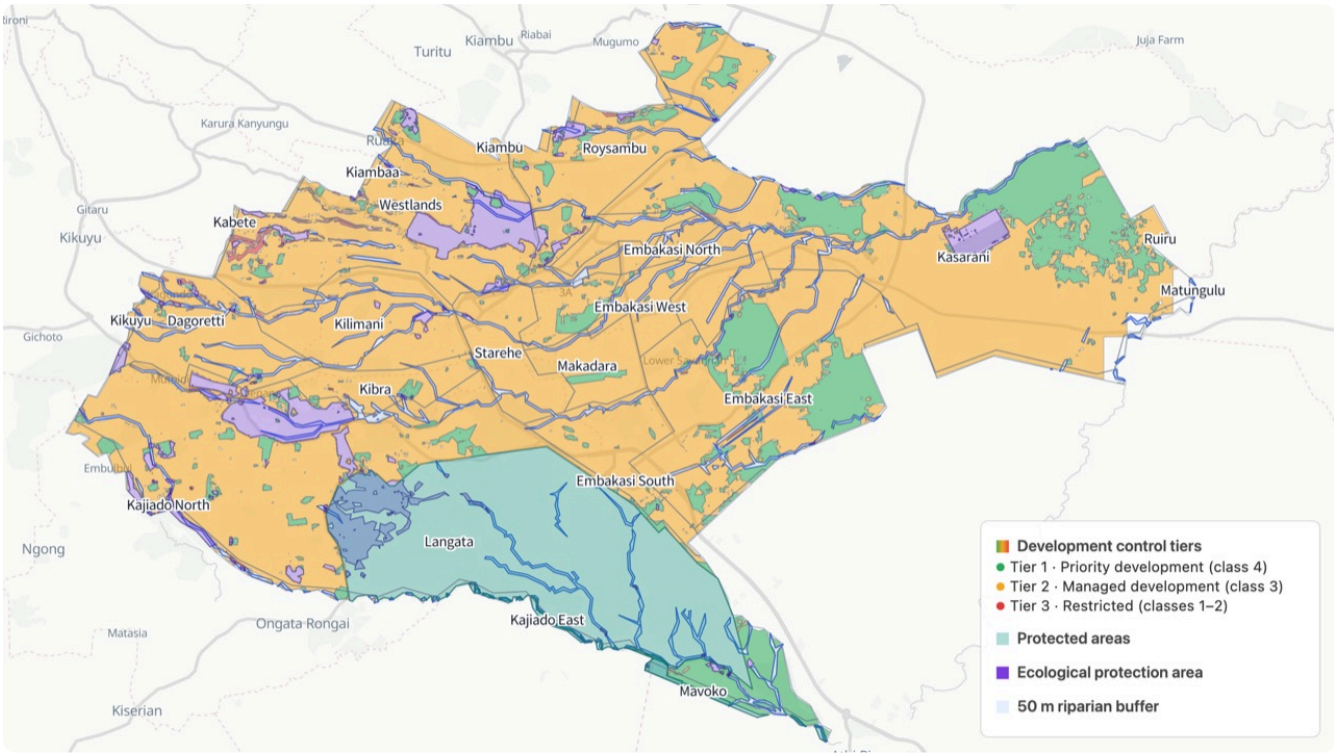
The layer names follow the run language, but the deliverable set is stable enough to plan a handover around.

> Open **15-minute Neighbourhood Coverage** from the scenario catalog at **app.mapxagent.com** and start from its suggested question.

SCENARIO 02

Development Suitability Baseline Nairobi

Screen ecological, terrain and flood constraints, grade the remaining land, estimate carrying capacity and propose a development boundary with tiered controls.



The Nairobi scenario as delivered: development control tiers drawn over ecological protection areas, protected areas and the 50 m riparian buffer

AT A GLANCE

STUDY AREA **Nairobi**	ENTRY LAYERS **8 layers**
KEY METHODS **Hard-constraint screening · weighted overlay · capacity estimate**	DELIVERABLES **Suitability classes, constraint layers, capacity tables, charts and a report**

Development Suitability Baseline answers the first question of any growth plan: **where can this city build, and how much can it hold** once ecological, terrain and flood constraints are respected. It follows the two-part appraisal logic used in Chinese territorial planning — hard constraints first, suitability and carrying capacity second, boundary and tiered controls last.

What the scenario ships

Opening the scenario imports eight prepared layers for the Nairobi window. Every layer comes from the **built-in datasets** catalog with its default style, legend, source, licence and attribution attached.

Layer	Data	Role in the study
Road network	52,642 OSM lines	Existing urban skeleton; reference for the boundary and for infill versus new growth
Watercourses	718 OSM lines	Source of the 50 m riparian buffer
Administrative boundaries	26 ADM3 units (geoBoundaries)	Attribution, summary tables and map labels
Protected area constraints	379 Overture features across Kenya	Part of the ecological exclusion
DEM	Copernicus DEM GLO-30, 30 m (1,459–1,954 m)	Slope classes
Land cover	Esri Sentinel-2 2020, 10 m, 9 classes (57% built-up)	Ecological classes and the existing built-up surface
Flood hazard	WRI Aqueduct, 1 km	Flood avoidance line
Population	GHS-POP, 1 km	Current population reference for the capacity comparison

*All layers come from the **built-in catalog** with source and attribution attached, and they are not limited to Nairobi: the catalog provides the same datasets for other cities at the same specification — switch the study area, add the same entries, and this chapter's method chain transfers in minutes.*

The layer order is the display order: points, then lines, then polygons, then rasters. All eight stay in the project for follow-up questions; only the surfaces you see in the legend above are styled as deliverables.

How the study works

The method is a chain of three planning questions. Each one maps onto deterministic **analysis** tools, so every number in the report can be traced back to a run.

Stage	Question	What MapX runs	Output
1. Screen the hard constraints	Which land is off-limits?	Slope, Reclassify, Buffer, Raster calculator, Overlay erase, Raster class area, Zonal statistics	Ecological protection mask and its polygons, the 50 m riparian buffer, constraint-area tables
2. Grade and size what remains	How suitable is the remaining land, and how many people can it carry?	Weighted overlay (equal weights by default), Reclassify, Grid at 1 km, Zonal statistics	Suitability classes 1–4, population per evaluation cell, class-area and capacity tables
3. Propose the boundary	Where should the development boundary run, and what tiers apply?	Raster to polygons, Overlay erase and dissolve, Zonal summary, Label anchors	Development control tiers (priority / managed / restricted), charts and a written report

In short, the study first subtracts the no-build space — ecological protection, steep slopes and the riparian buffer — from the study area, then grades the remaining land into four suitability classes with a weighted overlay and estimates carrying capacity per evaluation cell, and finally vectorises the result into three development control tiers (priority, managed, restricted) with charts and a report. The three stages share one deterministic operator chain: each stage's output layer is the next stage's input.

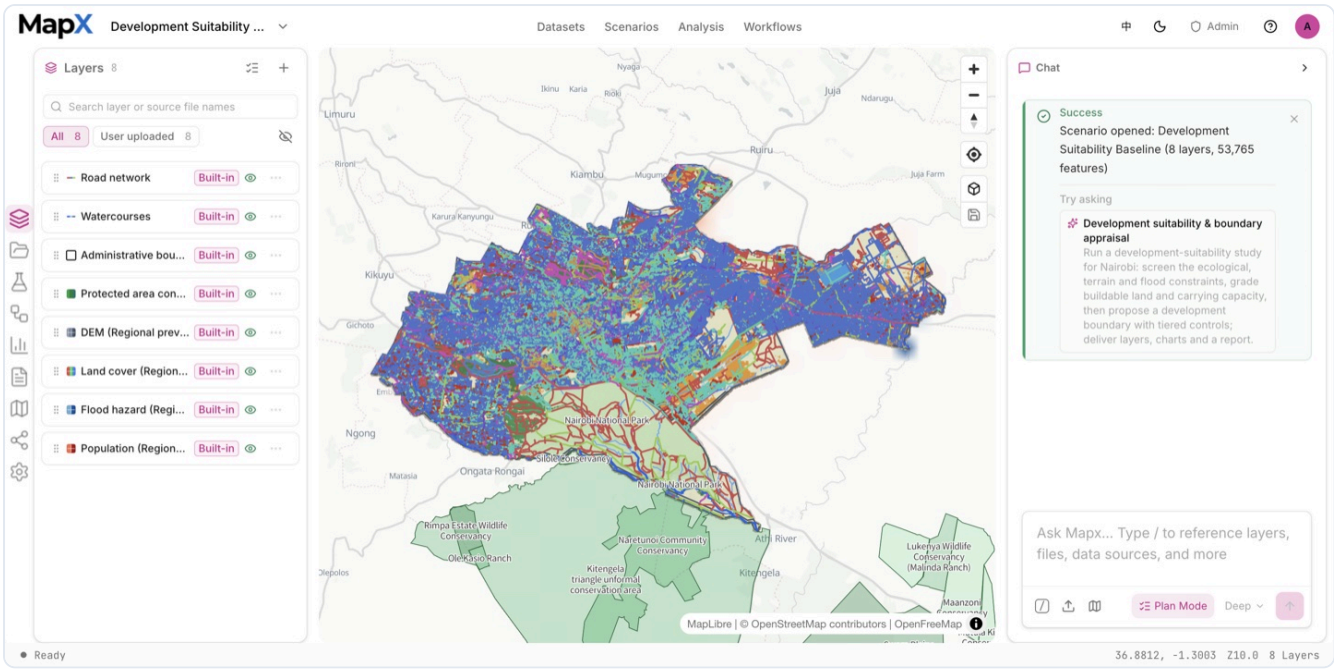
Run it

The screenshots below come from this scenario's reference workspace. Each step names the platform surface it uses, so you can follow the same path in your own project.

> **SUGGESTED QUESTION**
>
> *"Run a development-suitability study for Nairobi: screen the ecological, terrain and flood constraints, grade buildable land and carrying capacity, then propose a development boundary with tiered controls; deliver layers, charts and a report."*

1. Open the scenario

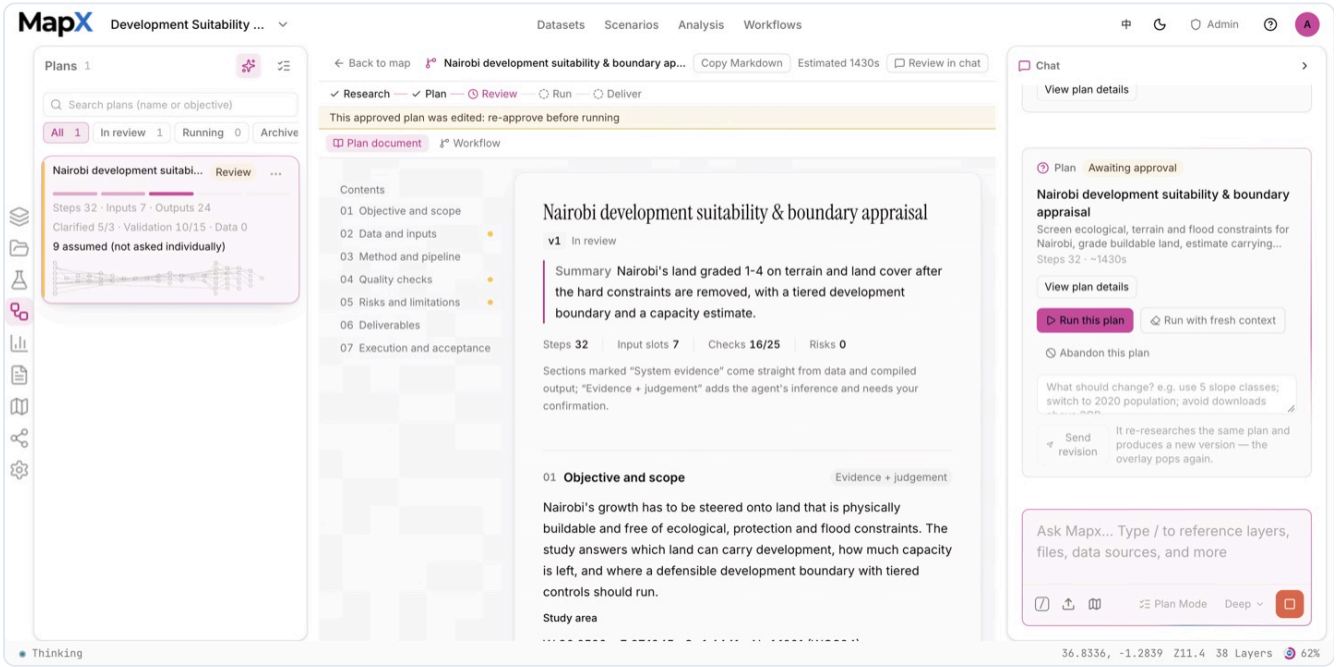
Open **Scenarios** in the top menu, select this card and choose **Open scenario**. The whole bundle imports at once: every layer keeps its default style, legend, source and attribution, and the camera flies to the study area. Use the **Layers** panel to switch layers on and off before you type anything.



The Nairobi scenario opened in a fresh project: the layer bundle imported, the Layers panel expanded, and the chat panel confirming the import with the suggested question

2. Ask the suggested question and review the plan

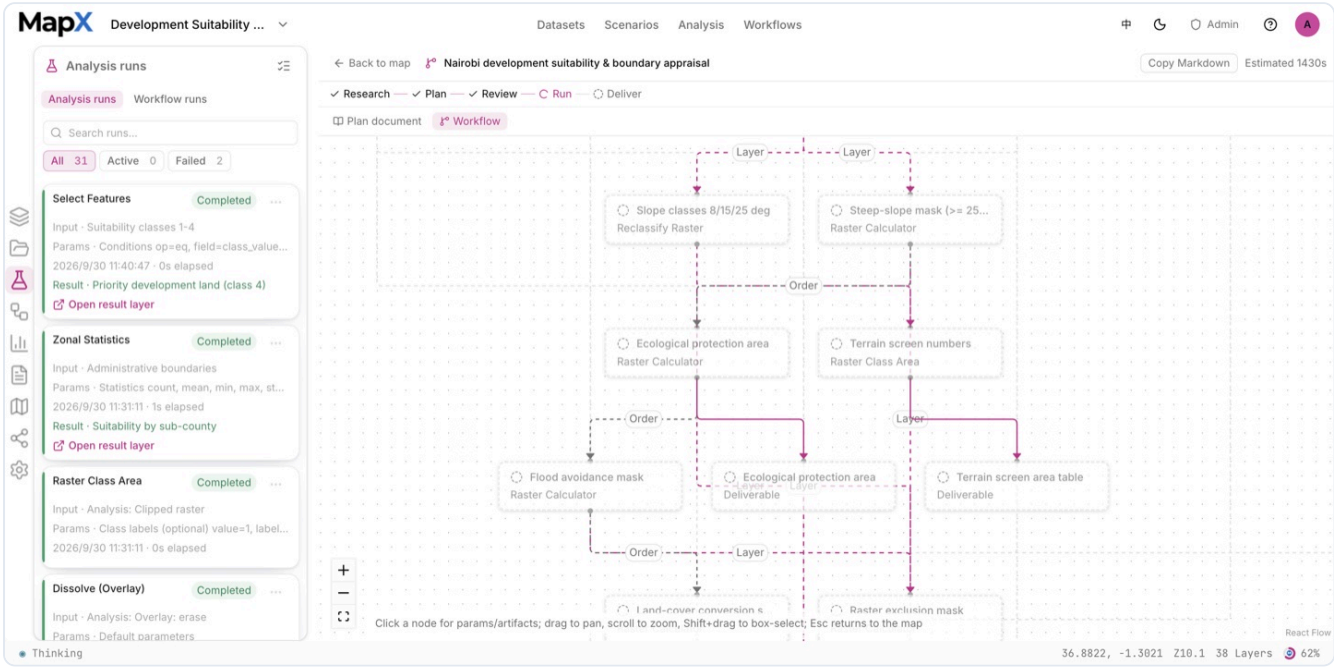
Send the **suggested question** in the box above — or edit it first. **Plan mode** turns it into a reviewable plan: the AI asks which admin units and constraints to use and whether the ecological mask is a hard exclusion, then writes the plan document — objective and scope, data inputs, method and pipeline, quality checks, risks, deliverables and execution. Every section is marked as **system evidence** or **evidence + judgement**, and the assumptions are listed before anything runs.



The Nairobi plan document with the Plans panel on the left and the plan awaiting approval in the chat panel; plan sections and recorded decisions readable

3. Approve the plan and run the analysis

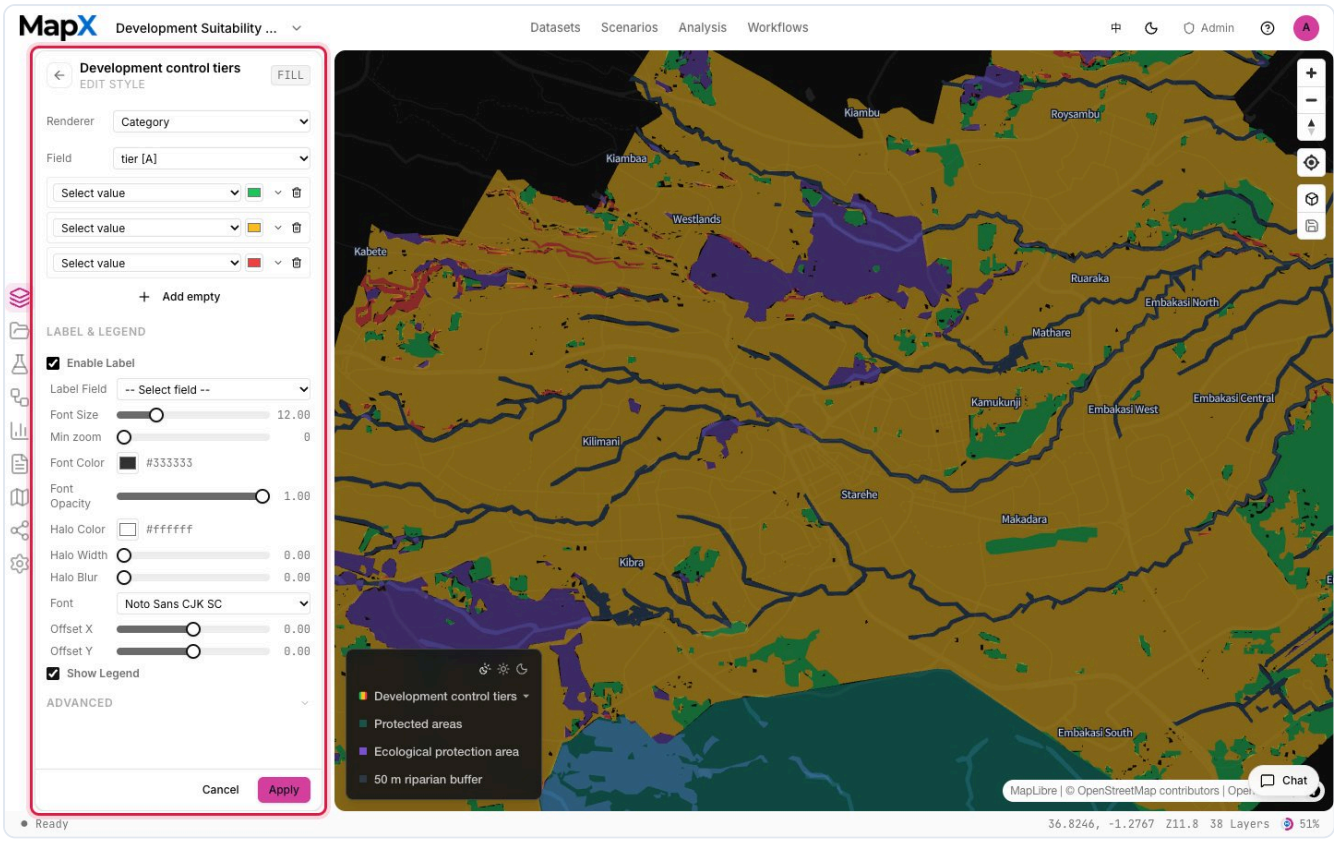
Approving the plan starts a chain of deterministic **analysis** tools: slope and reclassification, raster calculator masks for the ecological protection area, buffering along waterways, overlay erase for the hard exclusions, the 1 km evaluation grid and zonal statistics. The **Analysis** panel keeps the operator, inputs, parameters, status and result layer of every run, and the workflow graph in the centre visualises the whole chain, so each number in the report can be traced back to a step.



The analysis view for Nairobi: the Analysis panel lists every operator run on the left, with the workflow graph in the centre

4. Style the result on the map

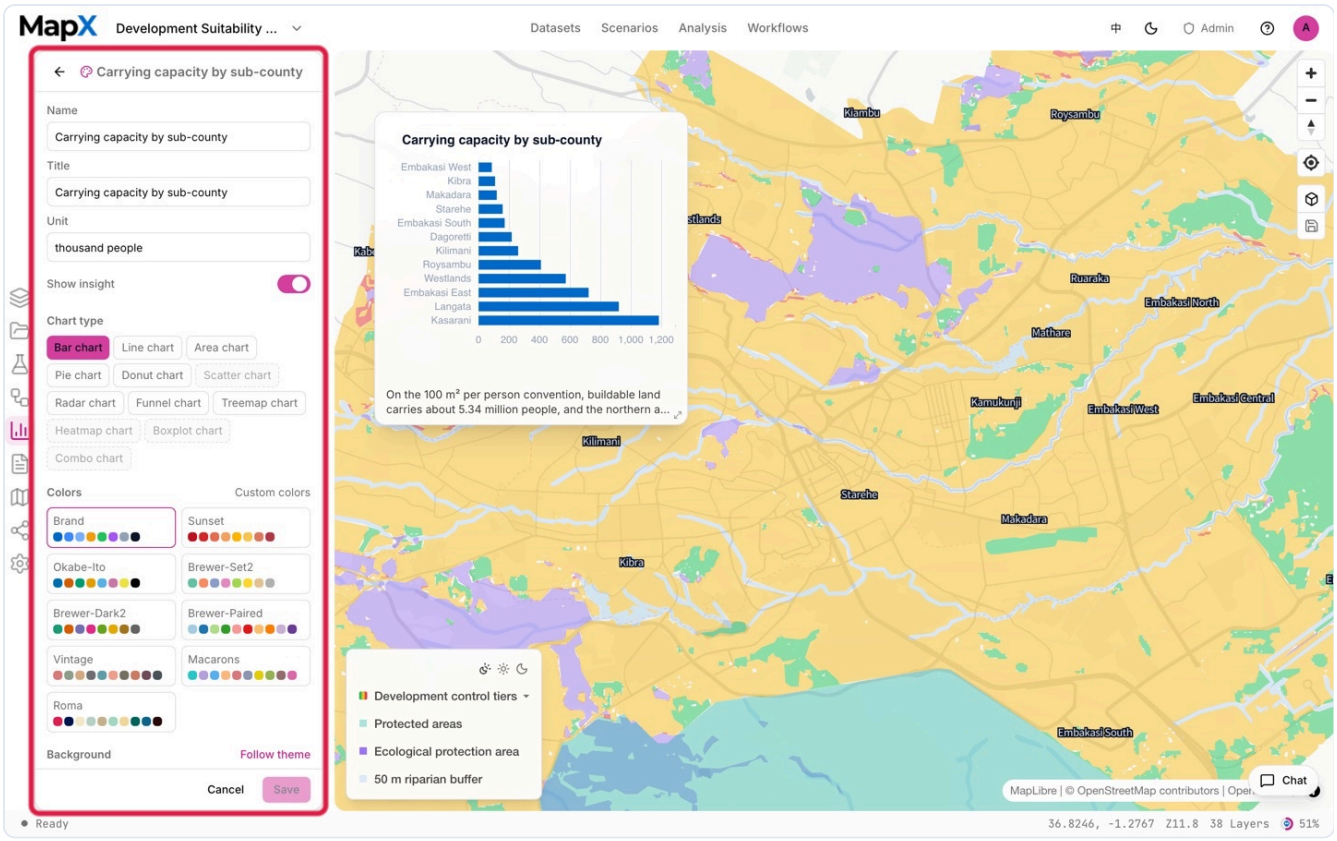
The layers arrive styled, and the **layer style editor** lets you change the renderer, colours, opacity, outline, labels and legend without writing code. Here the development control tiers keep human-readable class names, labels come from an anchor layer, and the constraint layers stay translucent so the tiers remain the strongest signal.



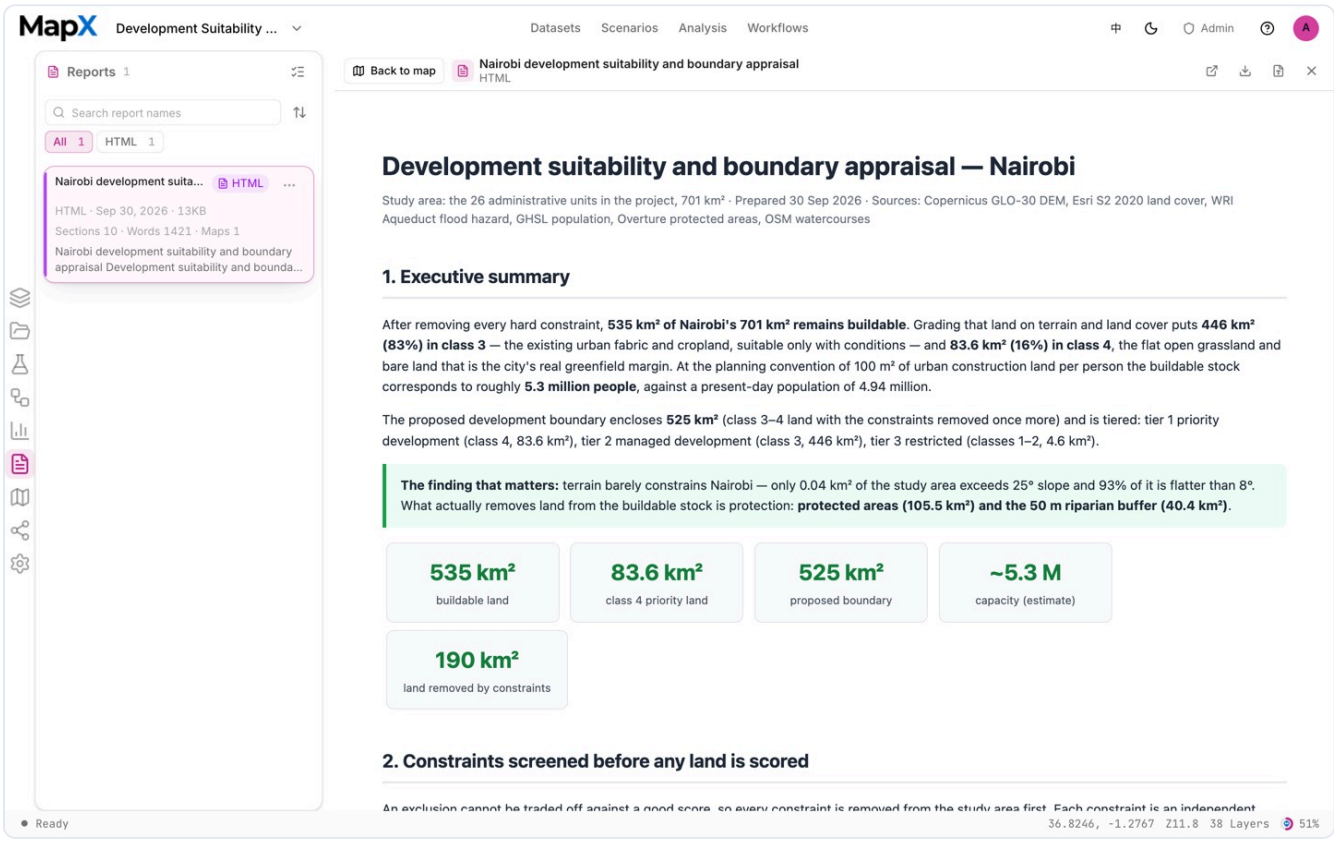
The style editor on a deliverable layer: classed-colour breaks and label settings, with the layer editing panel highlighted in red

5. Turn the result into charts and a report

Any result can become a **chart** — donut, pie, bar, heatmap, combo and more — with palettes, axis options and the choice to pin it to the map. The **report** canvas then combines the map, the charts and the tables into one document you can preview, revise and export; the same run can also produce a Word report for a formal submission.



A chart from this scenario in the chart editor, with the editing area highlighted in red



The scenario report opened in the report canvas, with an executive summary, figures and embedded maps

Example output

> **NOTE**
>
> The screenshots above come from one example run with the default settings on the shipped data. They show the shape of the deliverables, not what your run will return.

A finished run hands back a named, reusable set of deliverables instead of a single map:

- **Layers** — the buildable-land result, the suitability classes and the development control tiers, plus the constraint layers (ecological protection, protected areas, riparian buffer) as styled, legend-ready map layers.
- **Tables** — class areas with an admin cross table, the population per evaluation cell, and the district-level summary.
- **Charts** — the class-area composition, the constraints screened and the capacity ranking, each editable and pinnable.
- **Report** — an assessment document that combines the map, the figures and the threshold table, ready to share or export.

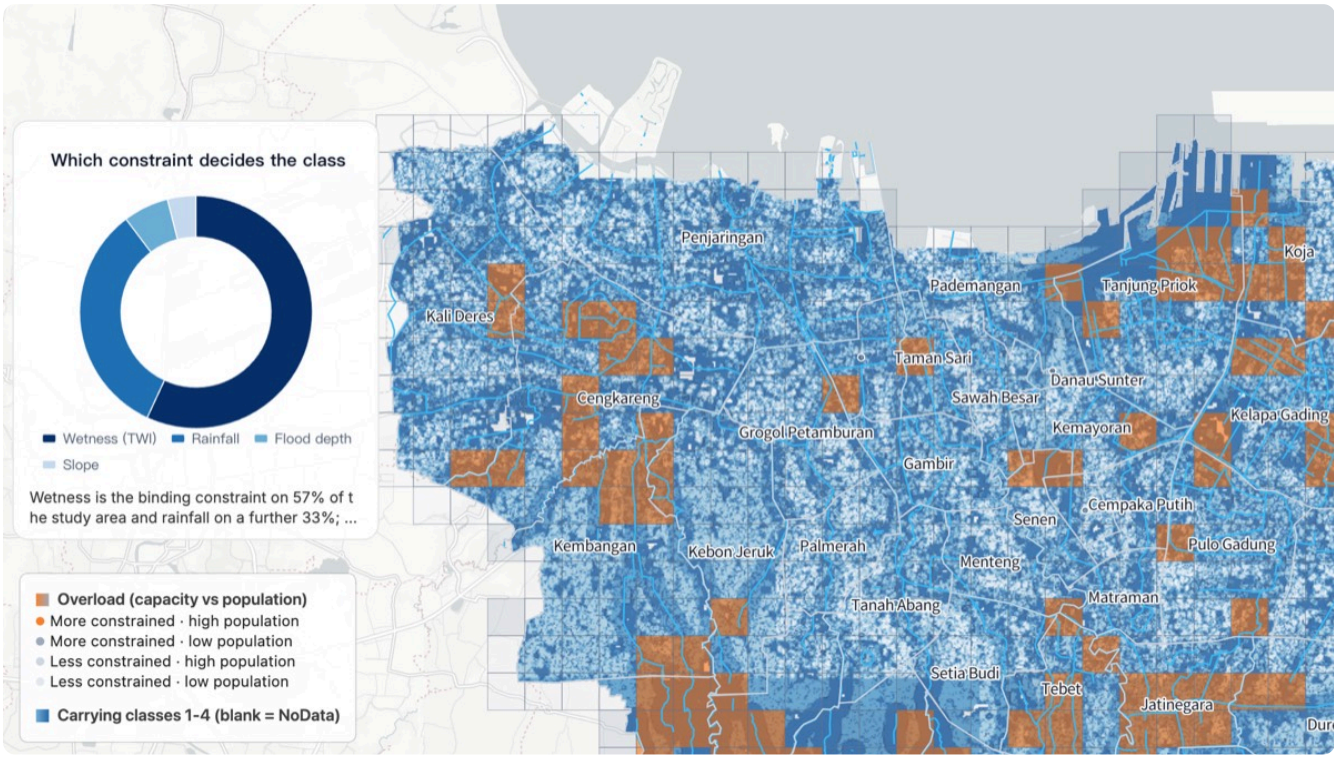
The layer names follow the run language, but the deliverable set is stable enough to plan a handover around.

Open **Development Suitability Baseline** from the scenario catalog at **app.mapxagent.com** and start from its suggested question.

SCENARIO 03

Carrying Capacity Assessment Jakarta

Assess how terrain, wetness, flood depth and rainfall constrain Jakarta's growth, grade carrying capacity, and flag the districts where pressure is highest.



The Jakarta scenario as delivered: carrying-capacity classes 1–4, overload cells and the constraint-composition chart

AT A GLANCE

STUDY AREA **Jakarta**	ENTRY LAYERS **6 layers**
KEY METHODS **Fuzzy membership · limiting-factor composite · pressure quadrants**	DELIVERABLES **Capacity classes, overload cells, constraint tables, charts and a report**

Carrying Capacity Assessment asks how much development a city can absorb before its water, climate and hazard constraints become the binding limit. For Jakarta — a low, flat delta city with frequent waterlogging — the study reads four constraints together, grades the result, and separates "constrained land" from "land under pressure".

What the scenario ships

Six prepared layers cover the Jakarta window.

Layer	Data	Role in the study
River network	OSM waterways	Drainage and flood corridor
Administrative boundaries	46 kecamatan (OCHA COD-AB ADM3); 44 on the mainland	Study-area definition, attribution and district labels
Precipitation	CHIRPS, ~5.35 km	Rainfall membership (waterlogging constraint)
Population	WorldPop, 100 m	Exposure and pressure axis
Flood hazard	WRI Aqueduct, ~900 m	Flood-depth membership
Terrain	SRTM, 30 m	Slope and topographic wetness

*All layers come from the **built-in catalog** with source and attribution attached, and they are not limited to Jakarta: the catalog provides the same datasets for other cities at the same specification — switch the study area, add the same entries, and this chapter's method chain transfers in minutes.*

How the study works

The design follows the "shortest stave" reading of carrying capacity: the most limiting factor decides the class, not the average.

Stage	Question	What MapX runs	Output
1. Build the capacity classes	Which factor limits each place, and how strongly?	Slope, Topographic wetness index, Fuzzy membership, Raster calculator (`minimum`), Focal statistics repair, Euclidean distance, Reclassify	Four-class carrying-capacity raster with the 300 m drainage-corridor adjustment
2. Diagnose pressure	Where is low capacity combined with high population?	Grid at 1 km, Zonal statistics, Quadrant (median split)	Overload cells, population per cell, district pressure table
3. Explain and mitigate	Which constraint dominates, and what should each district do?	Raster class area, Zonal statistics, Charts, Report	Class-area and dominant-constraint tables, mounted pressure chart and report

In short, the study first converts the four constraints — terrain, wetness, flood and rainfall — into comparable membership scores and composes them into capacity classes on a shortest-stave rule; it then crosses the classes with population in a quadrant read to find the overloaded cells where capacity is low and pressure high; finally it breaks down the dominant limiting factor per district and assembles the classes, overload grid and measures into charts and a report.

Run it

The screenshots below come from this scenario's reference workspace. Each step names the platform surface it uses, so you can follow the same path in your own project.

> **SUGGESTED QUESTION**
>
> *"Run a resource-and-environment carrying-capacity study for Jakarta: assess how water, climate and flood constraints limit growth, flag overloaded areas, then recommend measures by dominant constraint; deliver layers, charts and a report."*

1. Open the scenario

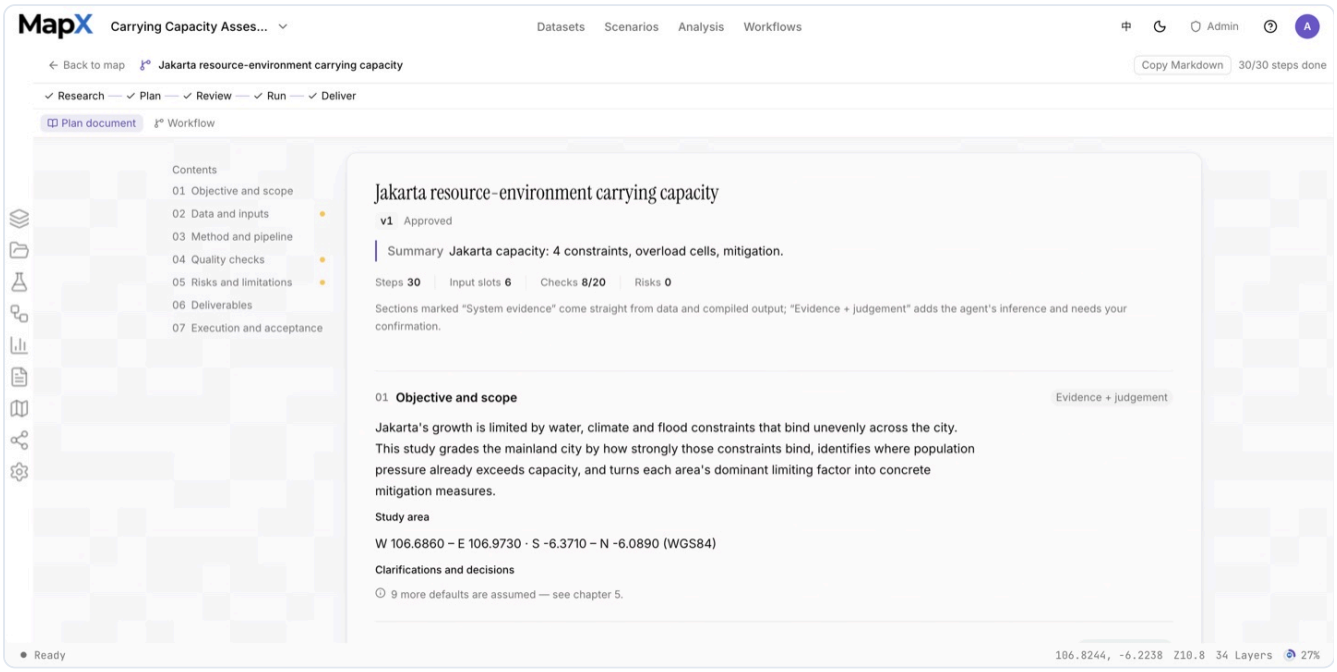
Open **Scenarios** in the top menu, select this card and choose **Open scenario**. The whole bundle imports at once: every layer keeps its default style, legend, source and attribution, and the camera flies to the study area. Use the **Layers** panel to switch layers on and off before you type anything.



The Jakarta scenario opened in a fresh project: the layer bundle imported, the Layers panel expanded, and the chat panel confirming the import with the suggested question

2. Ask the suggested question and review the plan

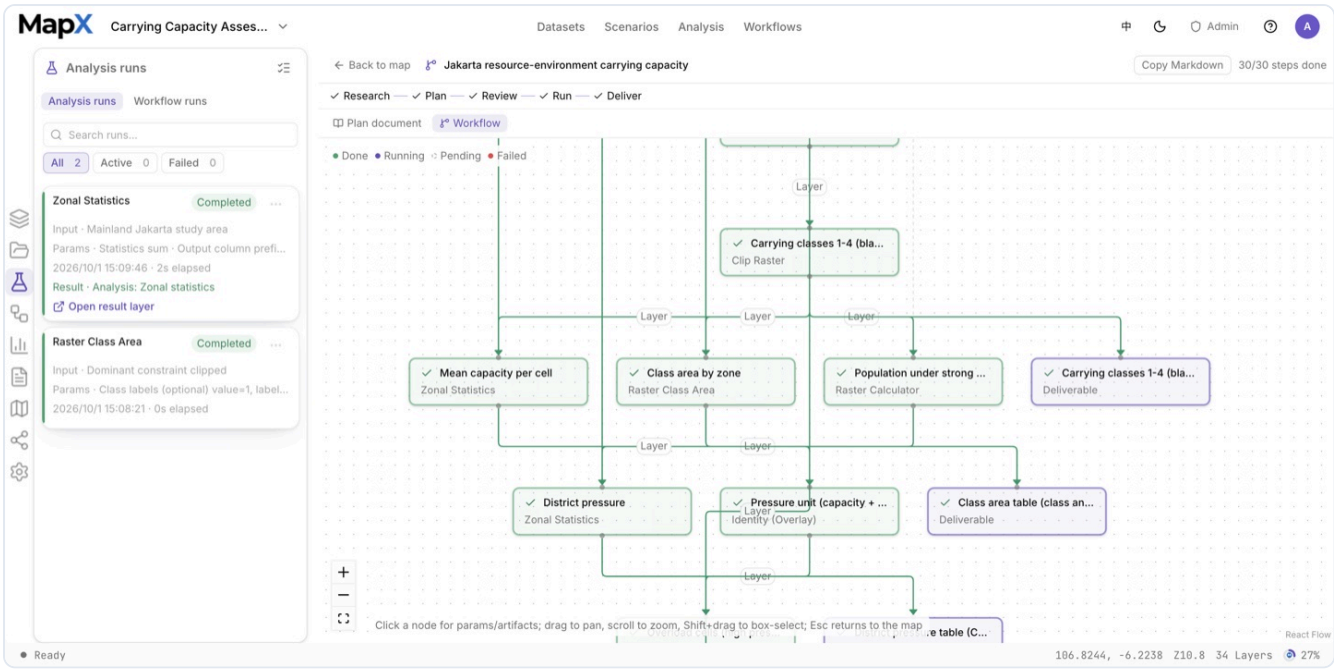
Send the **suggested question** in the box above — or edit it first. **Plan mode** asks how to read the study extent, whether to use the strict limiting-factor rule or the compensating variant, and which numbers the report should quote — then writes the plan document with its data inputs, method, quality checks, risks and deliverables.



The Jakarta plan document in full-width view: side panels collapsed, plan sections and recorded decisions readable

3. Approve the plan and run the analysis

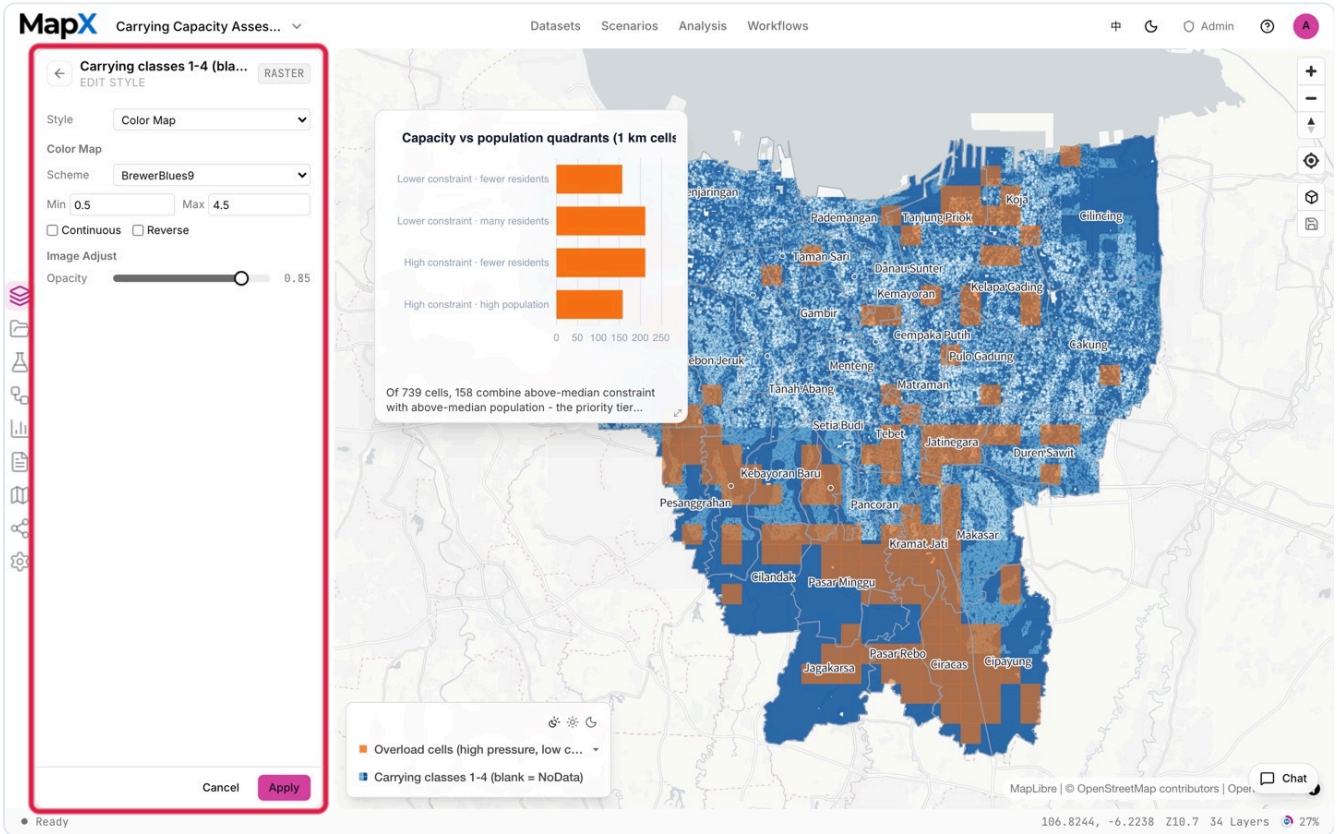
The run chains deterministic **analysis** tools: slope and topographic wetness, fuzzy membership for each limiting factor, focal-statistics repair for the two disclosed input gaps, the limiting-factor composite (shortest stave), the 300 m drainage corridor, the 1 km grid, zonal statistics and the pressure quadrant. The **Analysis** panel keeps every run's operator, inputs, parameters and result layer, and the workflow graph in the centre visualises the whole chain.



The analysis view for Jakarta: the Analysis panel lists every operator run on the left, with the workflow graph in the centre

4. Style the result on the map

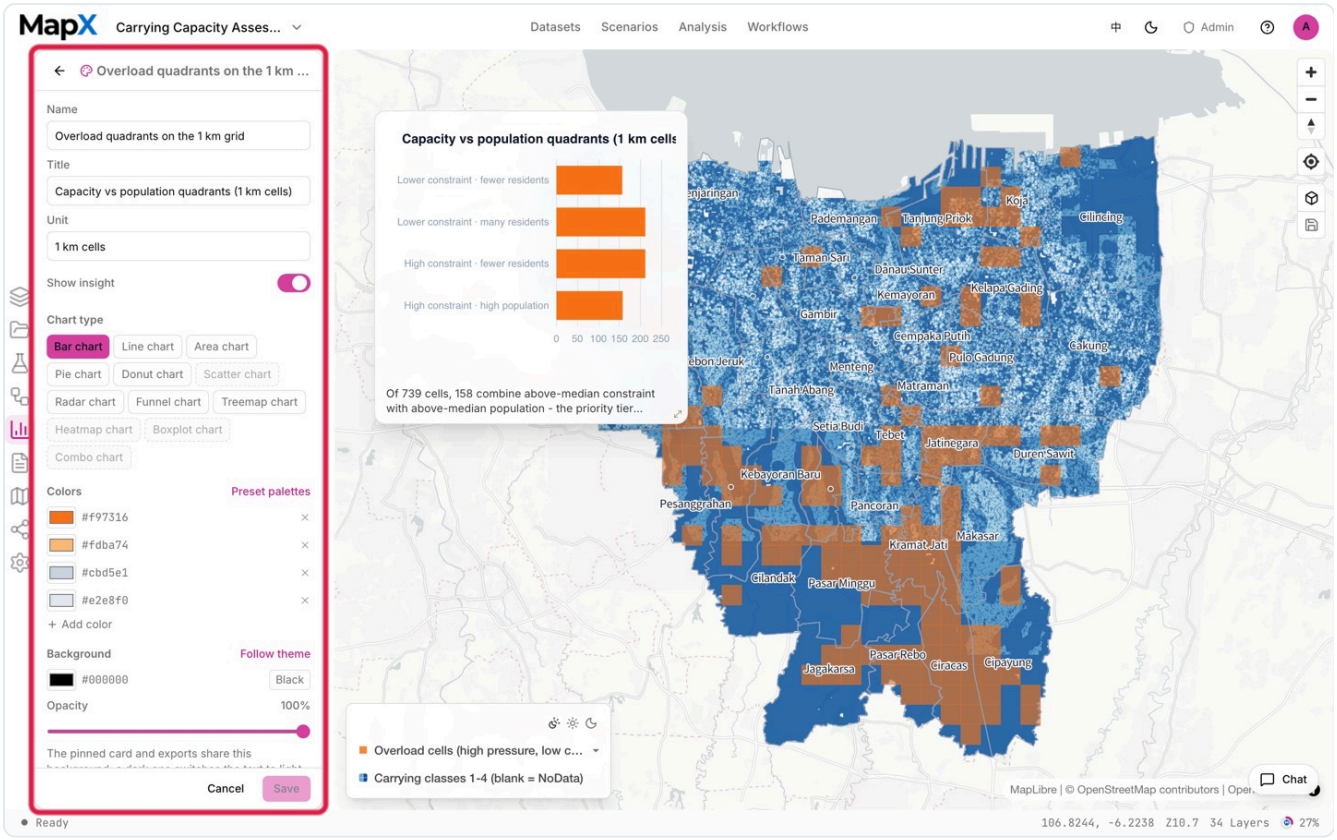
The carrying classes use a reversed blue ramp so the most constrained land is the darkest; the overload cells sit on top in orange, and district labels come from an anchor layer instead of the boundary polygons.



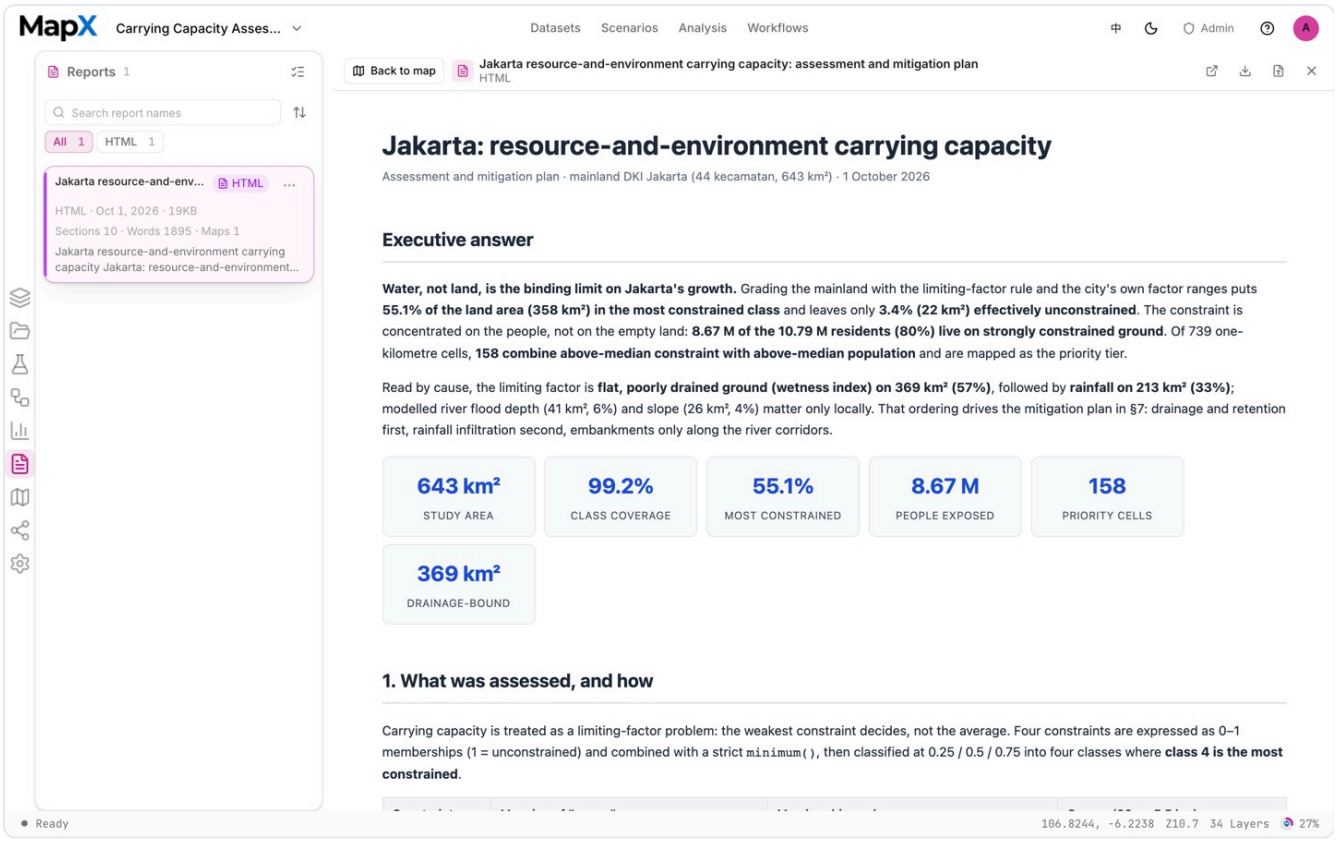
The style editor on a deliverable layer: classed-colour breaks and label settings, with the layer editing panel highlighted in red

5. Turn the result into charts and a report

The constraint-composition chart answers which factor decides each class, and the district charts rank the pressure. Both are **charts** you can restyle or pin, and the **report** collects the threshold table, the coverage statement and the class shares in one document.



A chart from this scenario in the chart editor, with the editing area highlighted in red



The scenario report opened in the report canvas, with an executive summary, figures and embedded maps

Example output

> **NOTE**
>
> The screenshots above come from one example run with the default settings on the shipped data. They show the shape of the deliverables, not what your run will return.

A finished run hands back a named, reusable set of deliverables instead of a single map:

- **Layers** — the four-class carrying-capacity raster with the drainage-corridor adjustment, the overload cells, the 1 km population grid and the district label layer.
- **Tables** — the class-area table with its admin cross table, the dominant-constraint table and the district pressure list.
- **Charts** — the constraint-composition chart and the pressure ranking, editable and ready to pin.
- **Report** — the assessment document with the threshold table, the data-repair statement and the five acceptance checks.

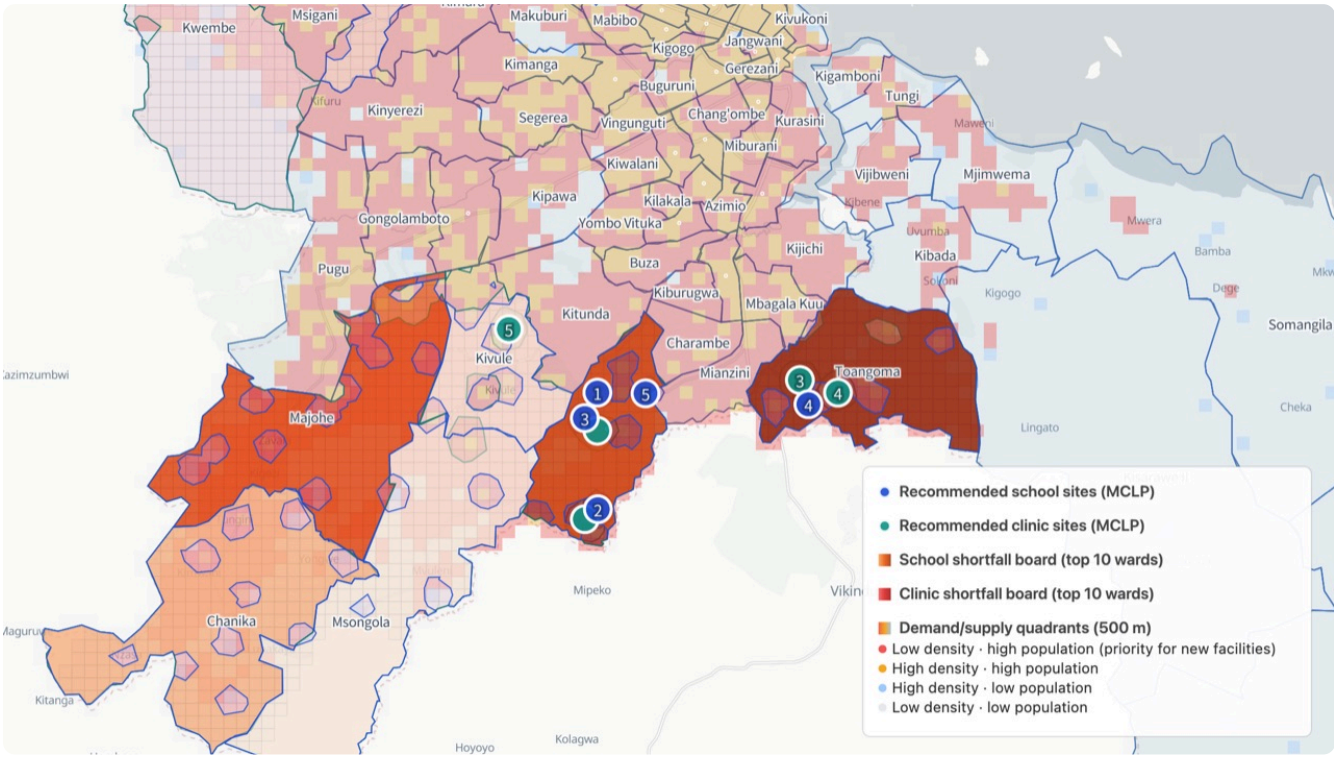
The layer names follow the run language, but the deliverable set is stable enough to plan a handover around.

Open **Carrying Capacity Assessment** from the scenario catalog at **app.mapxagent.com** and start from its suggested question.

SCENARIO 04

Public Facility Siting Dar es Salaam

Diagnose school and clinic coverage, measure the unserved demand, then compare candidate sites and recommend where to build with a maximum-coverage model.



The Dar es Salaam run as delivered: ward shortfall rankings, demand/supply quadrants and ranked MCLP sites for schools and clinics

AT A GLANCE

STUDY AREA **Dar es Salaam**	ENTRY LAYERS **7 layers**
KEY METHODS **Coverage diagnosis · candidate screening · MCLP vs k-median**	DELIVERABLES **Shortfall rankings, candidate cells, recommended sites, a marginal curve and a report**

Public Facility Siting answers the last mile of public-service planning: **given what already exists, where should the next schools and clinics go?** The scenario runs the full arc — coverage diagnosis, demand gap, candidate screening and siting comparison — and reports how many currently unserved residents each recommended site would bring within reach.

What the scenario ships

Seven layers cover the Dar es Salaam region.

Layer	Data	Role in the study
Education facilities	OSM education POIs (2,223)	Raw supply for the school analysis
Health facilities	Healthsites.io (1,463)	Raw supply for the clinic analysis
Road network	OSM roads (102,464 lines)	Walking/driving network for service areas and site snapping
Watercourses	OSM waterways, 11,160 lines	30 m riparian setback applied to candidates
Ward boundaries	OCHA COD-AB ADM3 (90 wards, 1,667 km²)	Shortfall ranking, candidate scope and balance checks
Building footprints	Overture (481,681 polygons)	Built-up intensity screen for candidate cells
Population	WorldPop, 100 m	Demand and coverage weights

*All layers come from the **built-in catalog** with source and attribution attached, and they are not limited to Dar es Salaam: the catalog provides the same datasets for other cities at the same specification — switch the study area, add the same entries, and this chapter's method chain transfers in minutes.*

How the study works

Stage	Question	What MapX runs	Output
1. Coverage diagnosis	Who is outside the standard today?	Select features (class screen), Overlay clip, Service area (800 m schools / 1000 m clinics), Overlay erase, Zonal statistics, Top N	Ward shortfall layer and a top-10 ranking per class
2. Demand gap	Where is demand high and supply low?	Grid at 500 m, Zonal statistics, Quadrant (median split), 1 km demand grid	Pressure quadrants (diagnostic) and the unserved-population demand layer
3. Siting comparison	Which new sites serve the most people?	Grid inside the worst wards, Overlay erase (30 m corridor), Zonal statistics/summary, Centroid points, Location-covering (MCLP), Location-allocation (k-median)	Recommended sites, the efficiency comparison, marginal coverage curve and residual gaps

In short, the study first audits the existing schools and clinics against their service standards to locate shortfall wards and unserved population; it then screens candidate grid cells inside the worst wards for setback and build-out conditions; finally a maximum-coverage model picks the recommended sites that serve the most unserved residents, with a second model as an efficiency reference, delivering recommended sites, a marginal coverage curve and a report.

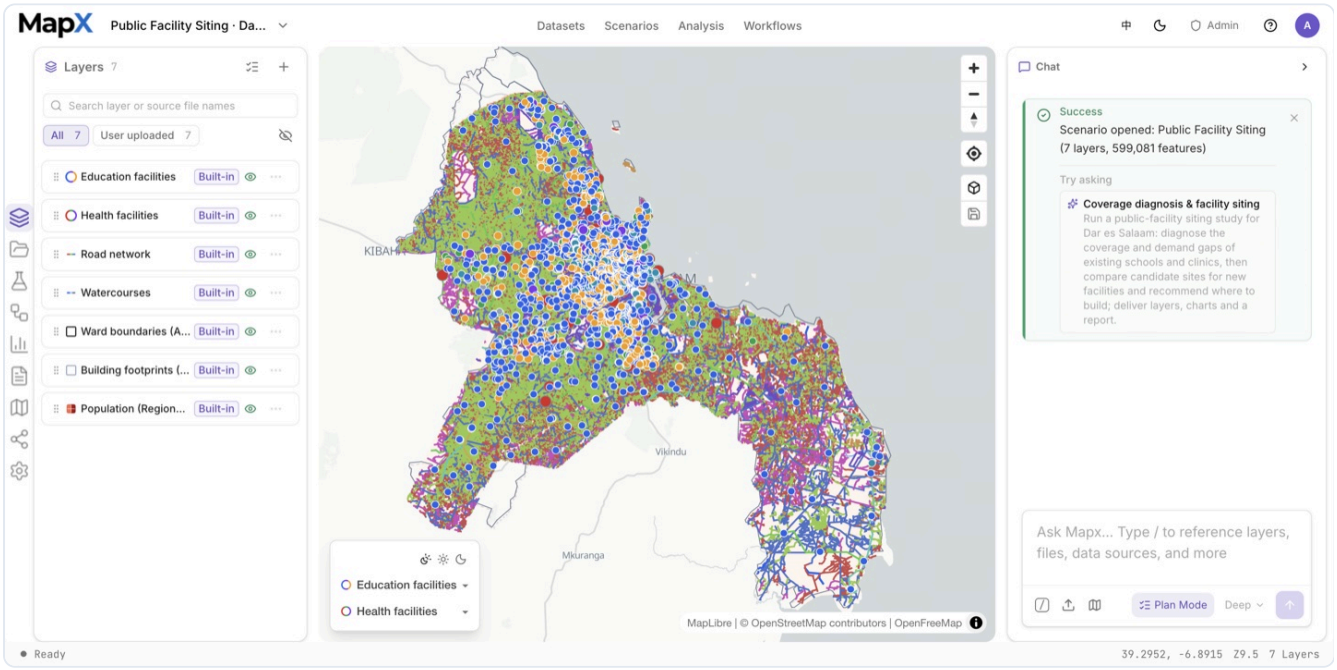
Run it

The screenshots below come from this scenario's reference workspace. Each step names the platform surface it uses, so you can follow the same path in your own project.

> **SUGGESTED QUESTION**
>
> *"Run a public-facility siting study for Dar es Salaam: diagnose the coverage and demand gaps of existing schools and clinics, then compare candidate sites for new facilities and recommend where to build; deliver layers, charts and a report."*

1. Open the scenario

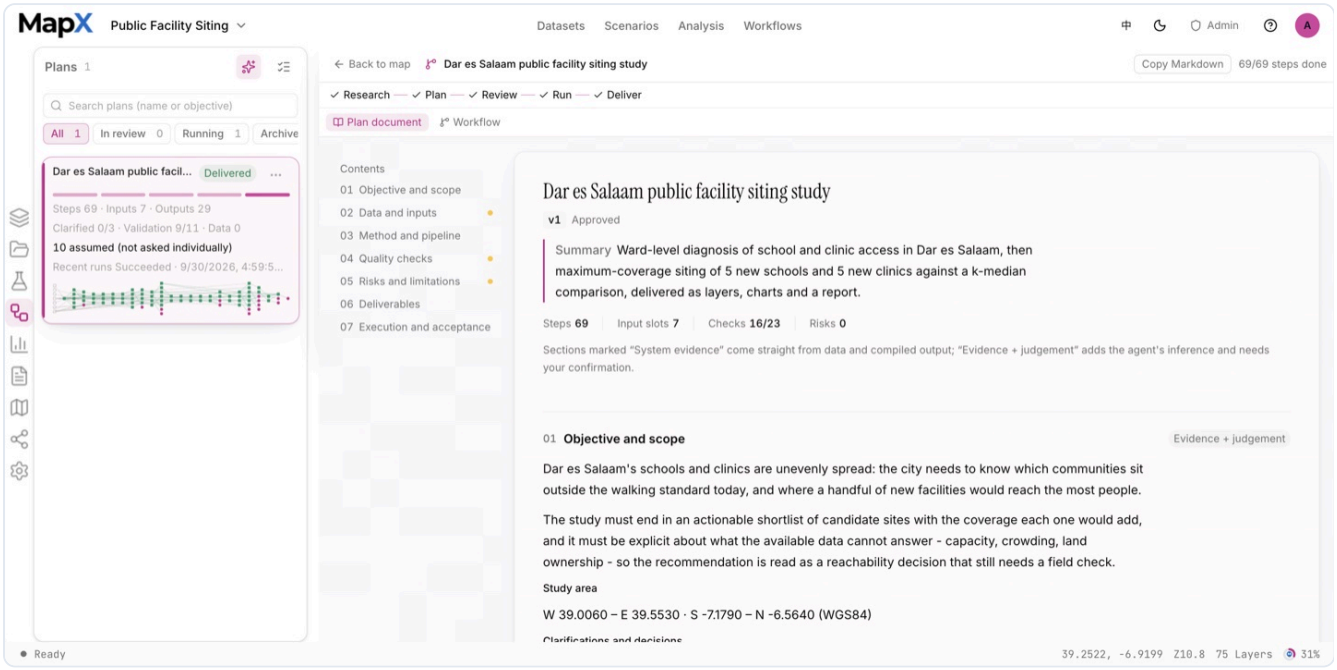
Open **Scenarios** in the top menu, select this card and choose **Open scenario**. The whole bundle imports at once: every layer keeps its default style, legend, source and attribution, and the camera flies to the study area. Use the **Layers** panel to switch layers on and off before you type anything.



The Dar es Salaam scenario opened in a fresh project: the layer bundle imported, the Layers panel expanded, and the chat panel confirming the import with the suggested question

2. Ask the suggested question and review the plan

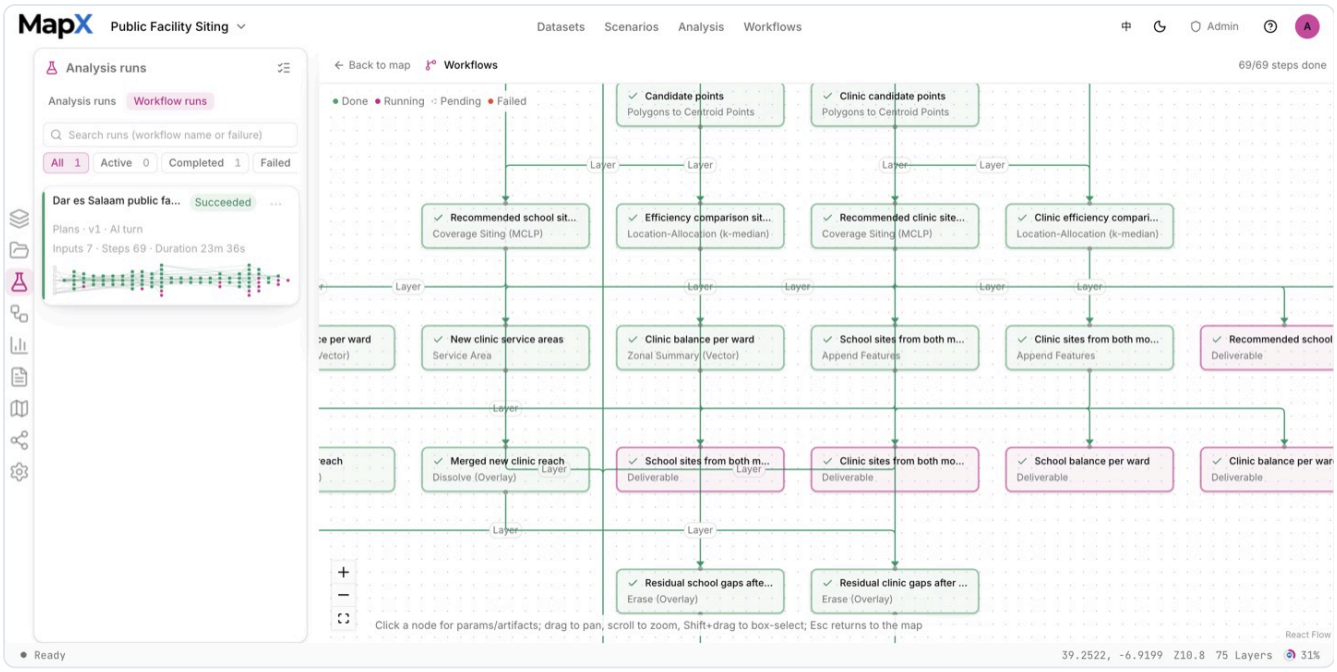
Send the **suggested question** in the box above — or edit it first. **Plan mode** records the supply-set definitions, the radius per class, the demand basis, the number of sites and the report language before the run starts.



The Dar es Salaam plan document with the Plans panel open on the left; plan sections and recorded decisions readable

3. Approve the plan and run the analysis

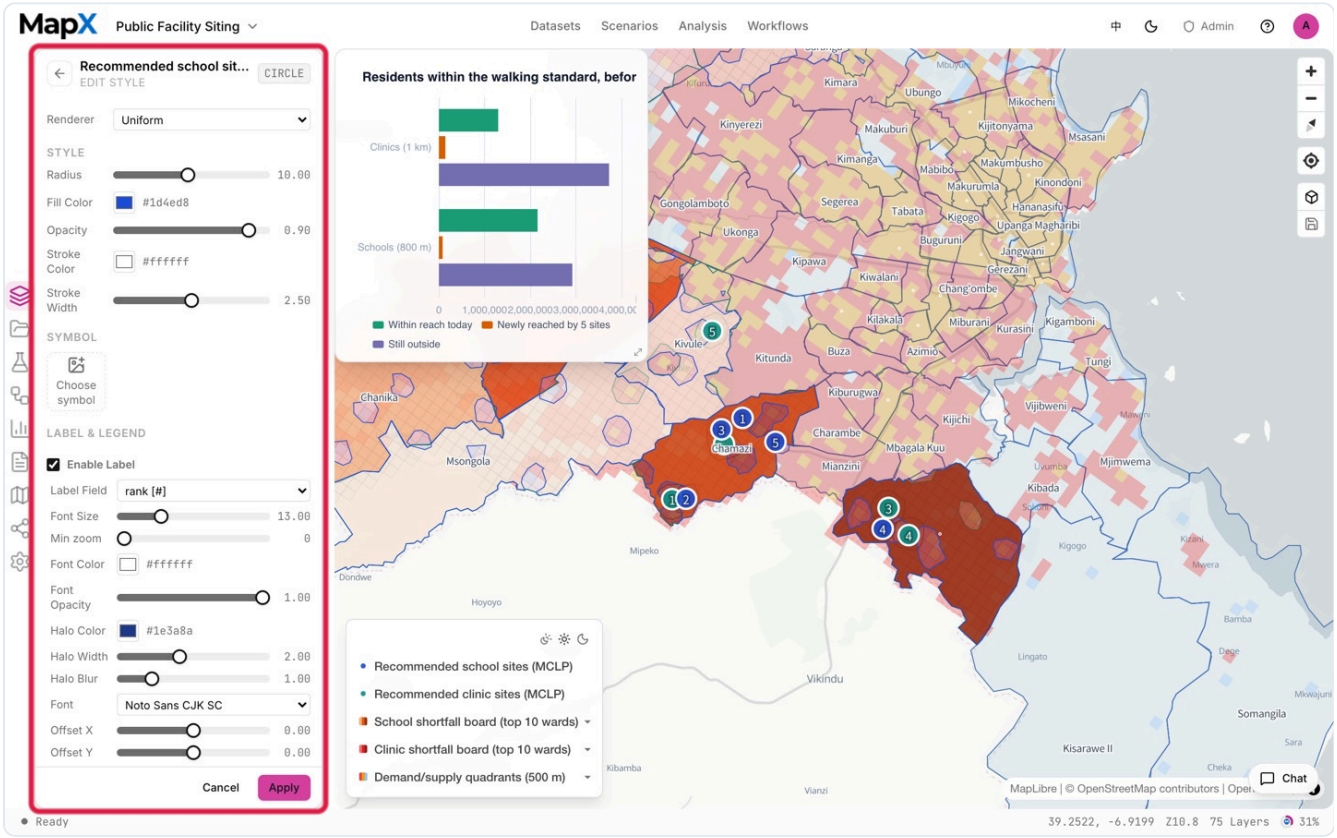
The chain screens the supply set by facility class, builds the status-quo **service areas** (800 m schools, 1000 m clinics), erases them from the wards to get the shortfall, prepares the unserved demand on a grid, screens the candidate cells and then runs the two siting models: maximum coverage (MCLP) as the recommendation and k-median as the efficiency comparison. The **Analysis** panel keeps every run's operator, inputs, parameters and result layer, and the workflow graph in the centre visualises the whole chain.



The analysis view for Dar es Salaam: the Analysis panel lists every operator run on the left, with the workflow graph in the centre

4. Style the result on the map

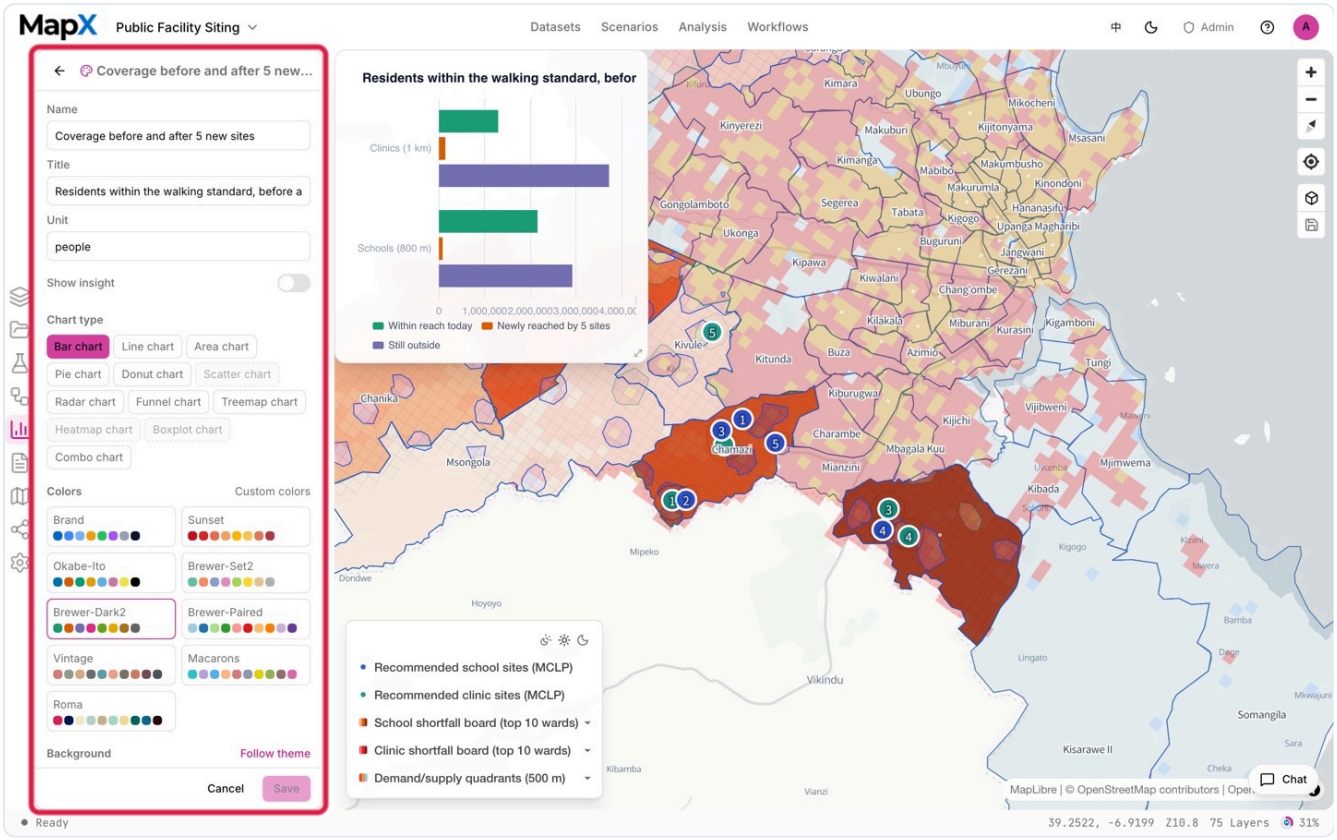
Recommended sites are ranked circles in the class hue, the shortfall rankings and demand/supply quadrants stay as hidden reference layers, and the legend states which model each marker belongs to.



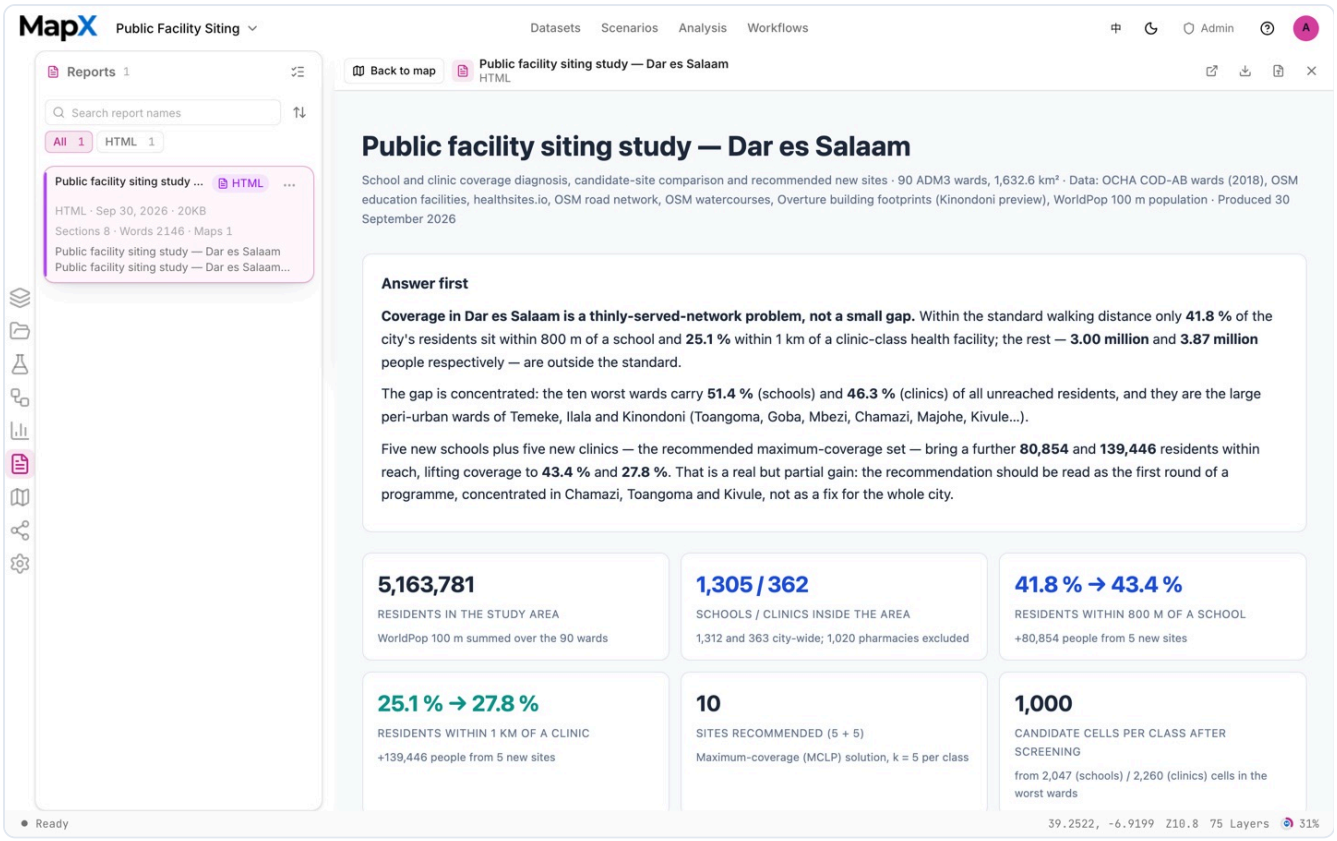
The style editor on a deliverable layer: classed-colour breaks and label settings, with the layer editing panel highlighted in red

5. Turn the result into charts and a report

The marginal coverage curve answers whether the next site still earns its place, and the before/after bars show what the recommended sites add. Both are **charts**; the **report** carries the before/after pair and every disclosure about capacity, crowding and fairness.



A chart from this scenario in the chart editor, with the editing area highlighted in red



The scenario report opened in the report canvas, with an executive summary, figures and embedded maps

Example output

> **NOTE**
>
> The screenshots above come from one example run with the default settings on the shipped data. They show the shape of the deliverables, not what your run will return.

A finished run hands back a named, reusable set of deliverables instead of a single map:

- **Layers** — the recommended sites for each facility class, the comparison sites, their service areas, the residual gaps, the shortfall rankings and the demand/supply quadrants.
- **Tables** — the ward balance table, the selected-site list and the feasibility counts for the candidate filters.
- **Charts** — the marginal coverage curve and the before/after coverage comparison.
- **Report** — the siting study with the supply-set definitions, the before/after pair and the capacity, crowding and fairness disclosures.

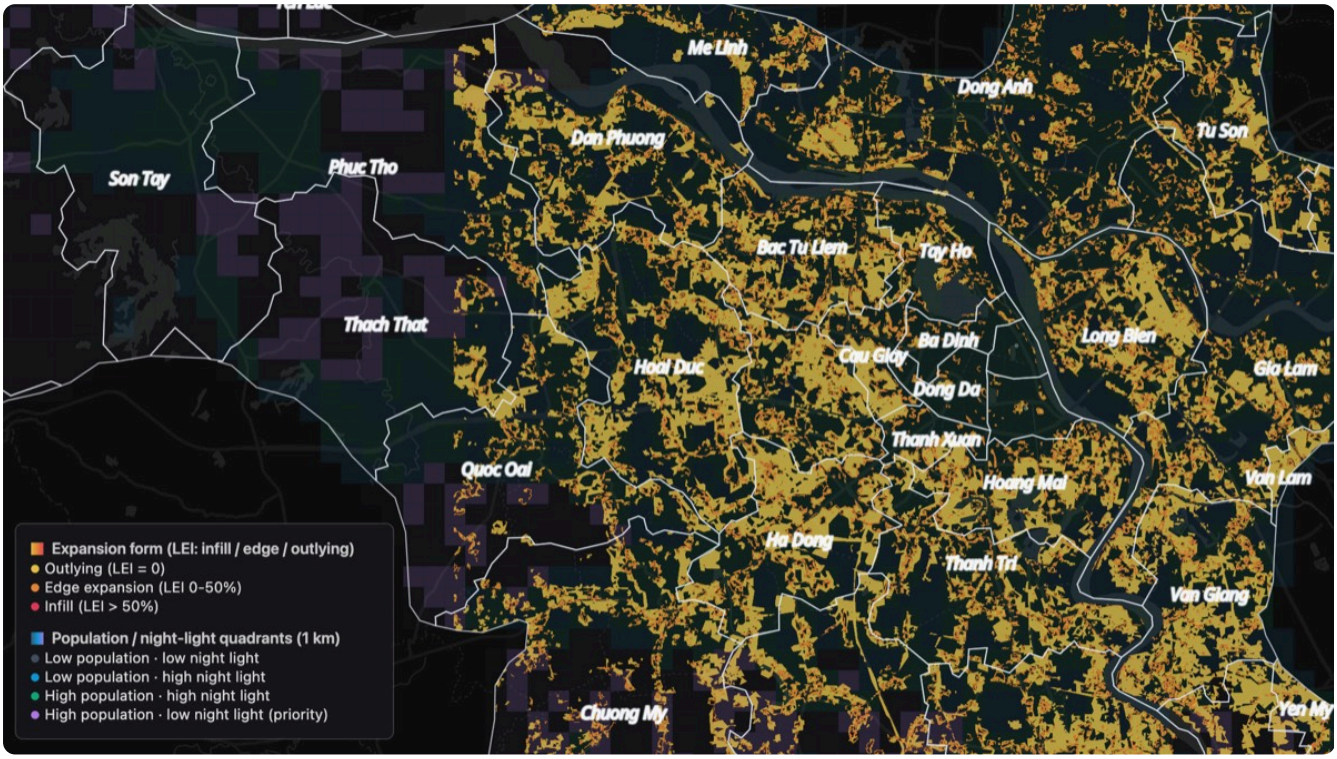
The layer names follow the run language, but the deliverable set is stable enough to plan a handover around.

Open **Public Facility Siting** from the scenario catalog at **app.mapxagent.com** and start from its suggested question.

SCENARIO 05

Urban Growth Monitor Hanoi

Reconstruct Hanoi's built-up expansion phase by phase, measure how much is infill versus outward growth, and connect growth to population and night lights.



The Hanoi run as delivered: infill, edge and outlying expansion forms over the built-up raster, with population and night-light quadrants

AT A GLANCE

STUDY AREA **Hanoi**	ENTRY LAYERS **7 layers**
KEY METHODS **Phased expansion metrics · landscape expansion index · centre and ellipse**	DELIVERABLES **Expansion-form layers, phase tables, quadrant grid, charts and a report**

Urban Growth Monitor asks three questions a growth-management plan needs to answer: **how fast is the city expanding, where is it going, and is the growth compact or sprawling?** It reconstructs the expansion phase by phase, separates infill from edge and outlying growth, and overlays population and night lights to find where growth and activity disagree.

What the scenario ships

Seven layers cover the Hanoi window. Only the built-up area is visible by default; the rest are context and analysis inputs.

Layer	Data	Role in the study
Road network	OSM roads	Test whether new built-up land follows the main axes
Watercourses	OSM waterways, 2,565 lines	River corridors as a growth constraint
Administrative boundaries	50 ADM3 units	Attribution and cross tables
Built-up area	GHSL 2020 density, max 7,967 m²/cell (visible)	Growth-intensity backdrop
Built-up history	WSF Evolution 1985–2015 build year, 733k valid cells	The phase and form source
Population	WorldPop 2020	Demand and activity axis
Night lights	VIIRS, December 2017 median, 611 m	Economic-activity proxy

*All layers come from the **built-in catalog** with source and attribution attached, and they are not limited to Hanoi: the catalog provides the same datasets for other cities at the same specification — switch the study area, add the same entries, and this chapter's method chain transfers in minutes.*

How the study works

Stage	Question	What MapX runs	Output
1. Reconstruct the expansion	How much built-up land was added in each period?	Expansion metrics (`periodBreaks` 1990 / 2000 / 2010 / 2015), Raster class area, Land change matrix	Phase table with new area, annual growth, intensity and compactness; built-year cross table
2. Measure the form	Is new growth infill, edge expansion or outlying?	Landscape expansion index on two built-up masks, Raster to polygons, Mean centre, Standard deviational ellipse, Directional distribution, Buffer and zonal summary	Infill / edge / outlying layer, centre shift and ellipse, share of new built-up near roads and waterways
3. Connect to activity	Where do population and night lights diverge?	Grid at 1 km, Zonal statistics, Quadrant (median split)	Population × night-light quadrants and the growth-management targets

In short, the study first reconstructs built-up expansion phase by phase, measuring new area, speed and intensity per period; it then classifies each new patch as infill, edge or leapfrog from how it touches the earlier built-up mask, and describes direction with the gravity centre and a standard deviational ellipse; finally it crosses growth with population and night lights in a quadrant grid to flag growth-management priorities, delivering phase layers, indicator tables and a report.

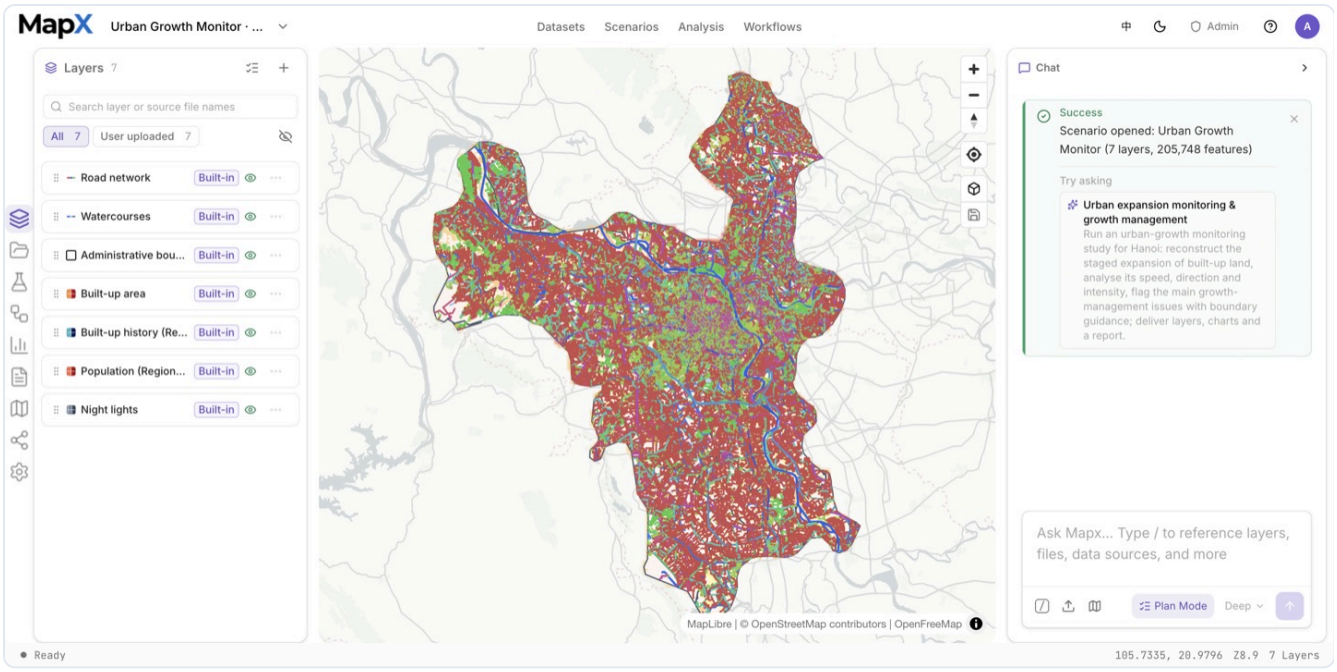
Run it

The screenshots below come from this scenario's reference workspace. Each step names the platform surface it uses, so you can follow the same path in your own project.

> **SUGGESTED QUESTION**
>
> *"Run an urban-growth monitoring study for Hanoi: reconstruct the staged expansion of built-up land, analyse its speed, direction and intensity, flag the main growth-management issues with boundary guidance; deliver layers, charts and a report."*

1. Open the scenario

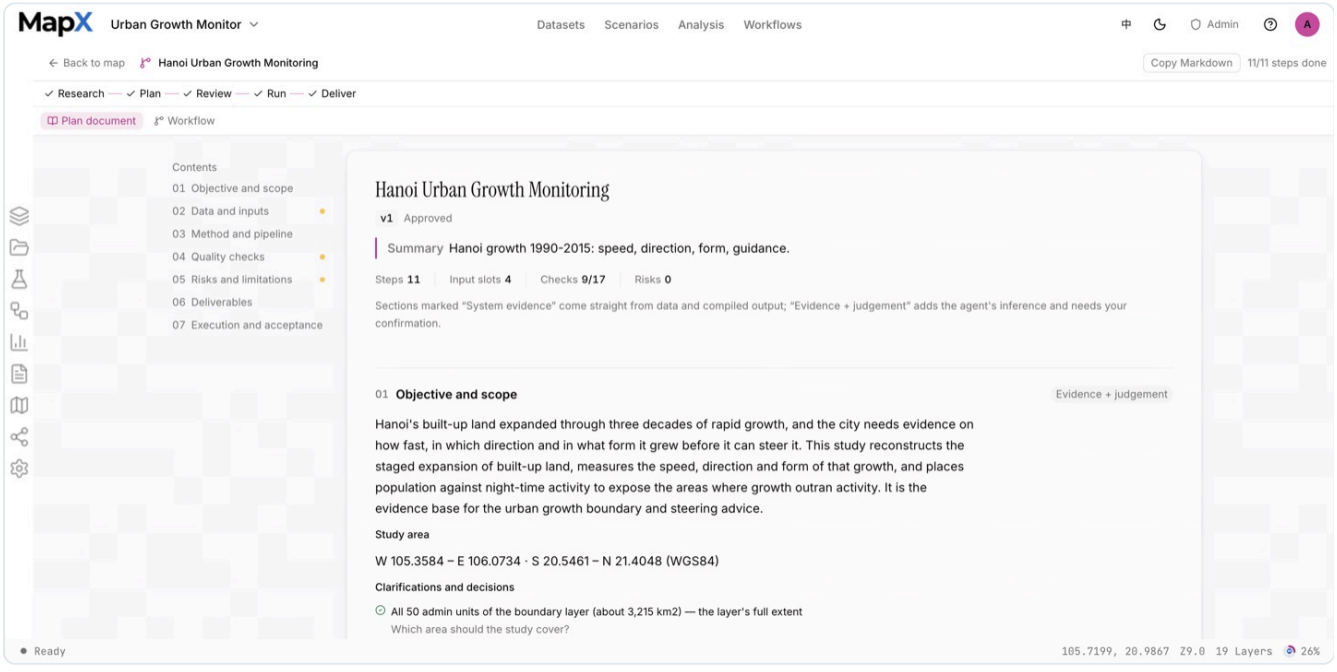
Open **Scenarios** in the top menu, select this card and choose **Open scenario**. The whole bundle imports at once: every layer keeps its default style, legend, source and attribution, and the camera flies to the study area. Use the **Layers** panel to switch layers on and off before you type anything.



The Hanoi scenario opened in a fresh project: the layer bundle imported, the Layers panel expanded, and the chat panel confirming the import with the suggested question

2. Ask the suggested question and review the plan

Send the **suggested question** in the box above — or edit it first. **Plan mode** confirms the study area, the two mask years used for the form measurement and the report numbers before anything runs.

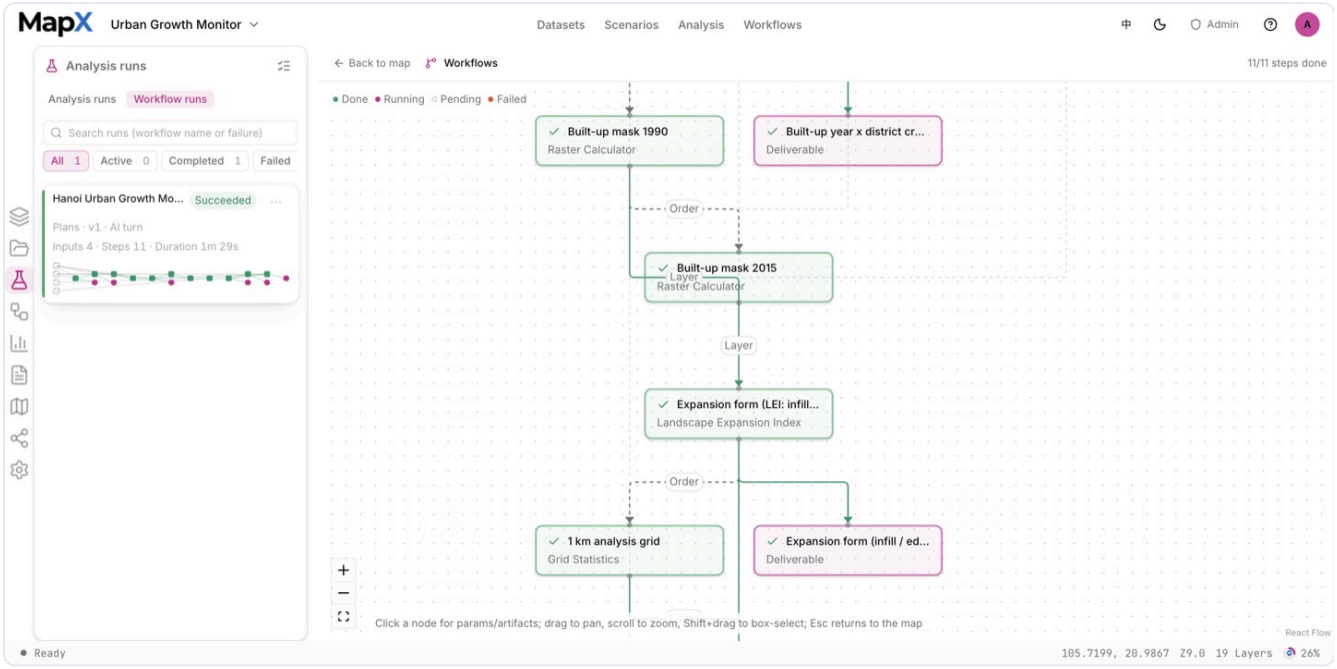


The Hanoi plan document in full-width view: side panels collapsed, plan sections and recorded decisions readable

3. Approve the plan and run the analysis

The run chains deterministic **analysis** tools: the phased expansion metrics, the landscape expansion index between the two built-up masks, polygon conversion for the form layer, the gravity centre and standard deviational ellipse, and the 1 km grid with population and night-light quadrants. The **Analysis** panel keeps

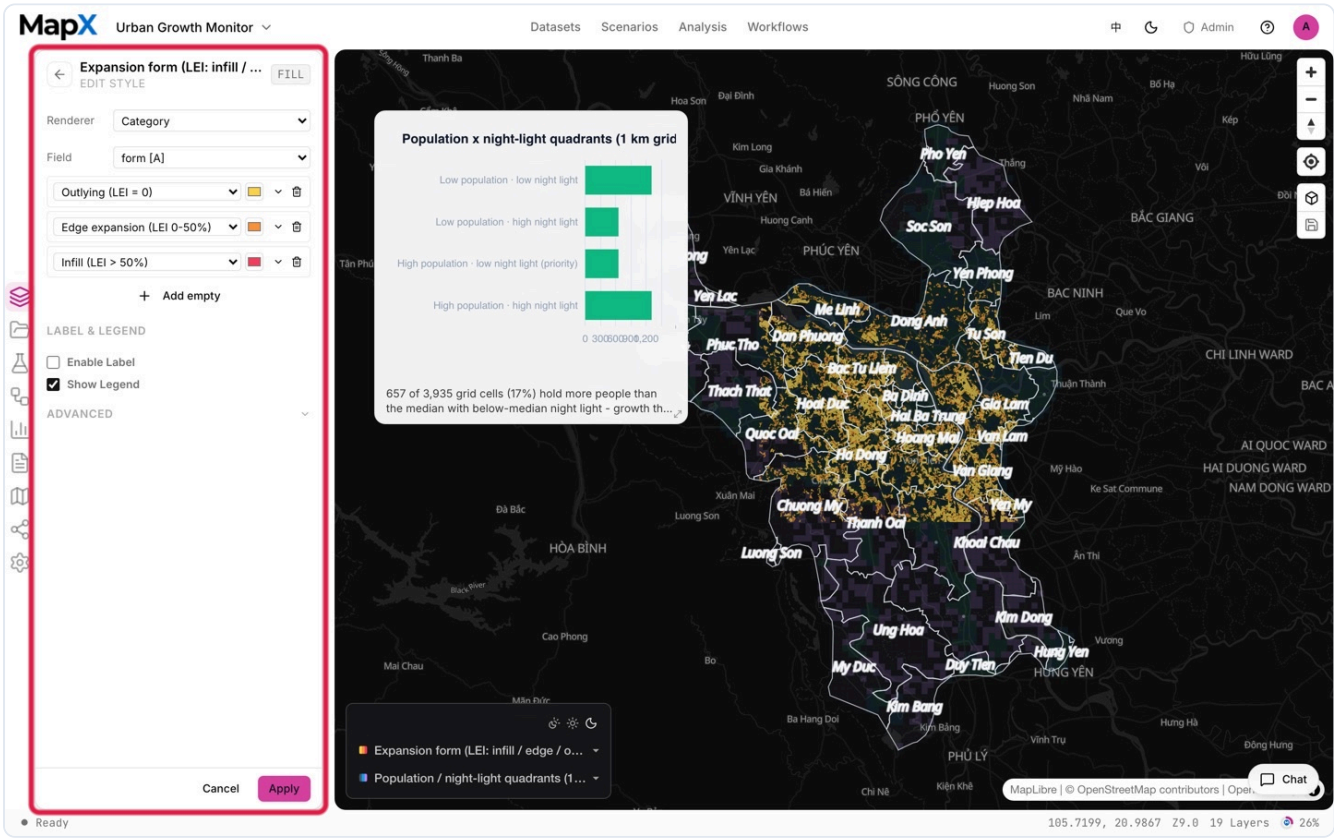
every run's operator, inputs, parameters and result layer, and the workflow graph in the centre visualises the whole chain.



The analysis view for Hanoi: the Analysis panel lists every operator run on the left, with the workflow graph in the centre

4. Style the result on the map

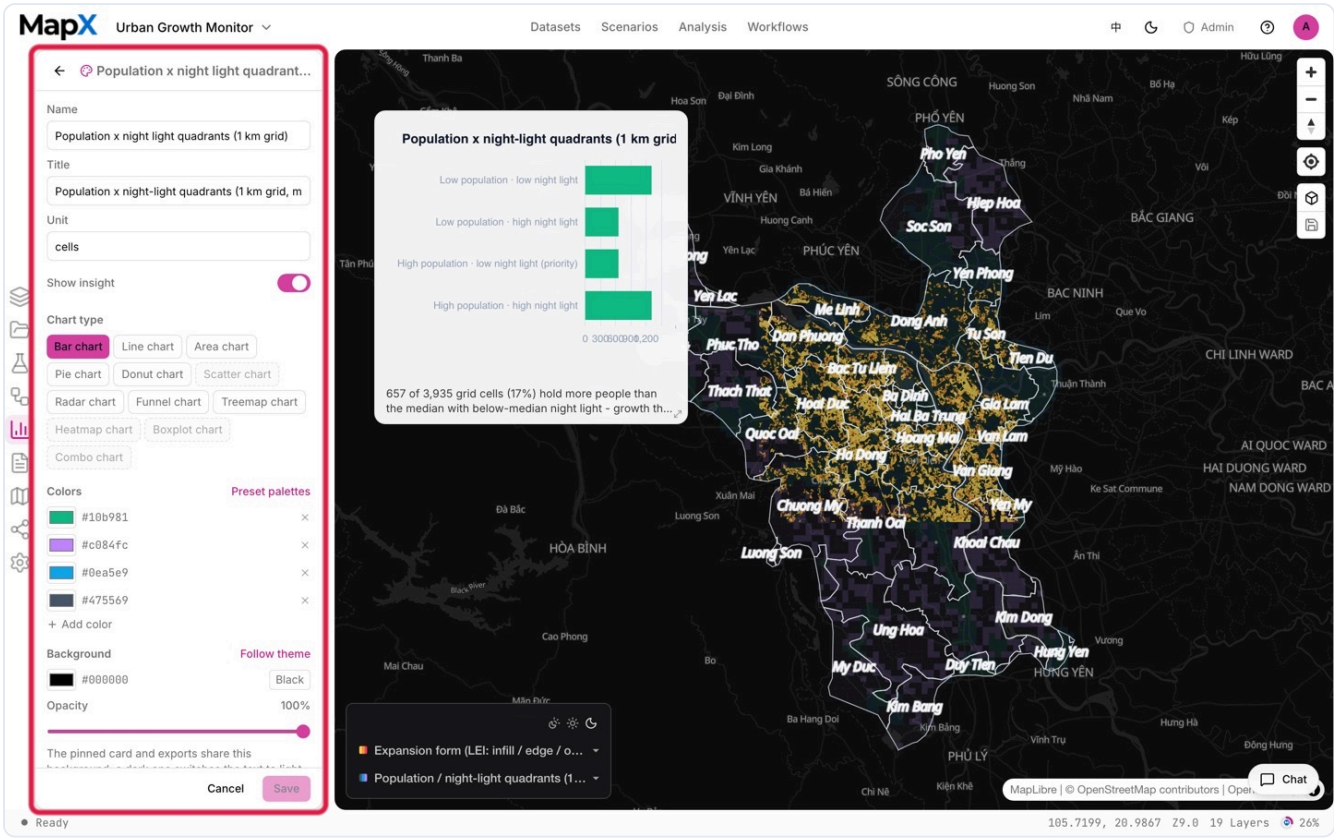
The expansion-form layer keeps three human-readable classes on the dark basemap, with the quadrant grid and district labels as light references; the layer style editor is where the hue family and label halo are set.



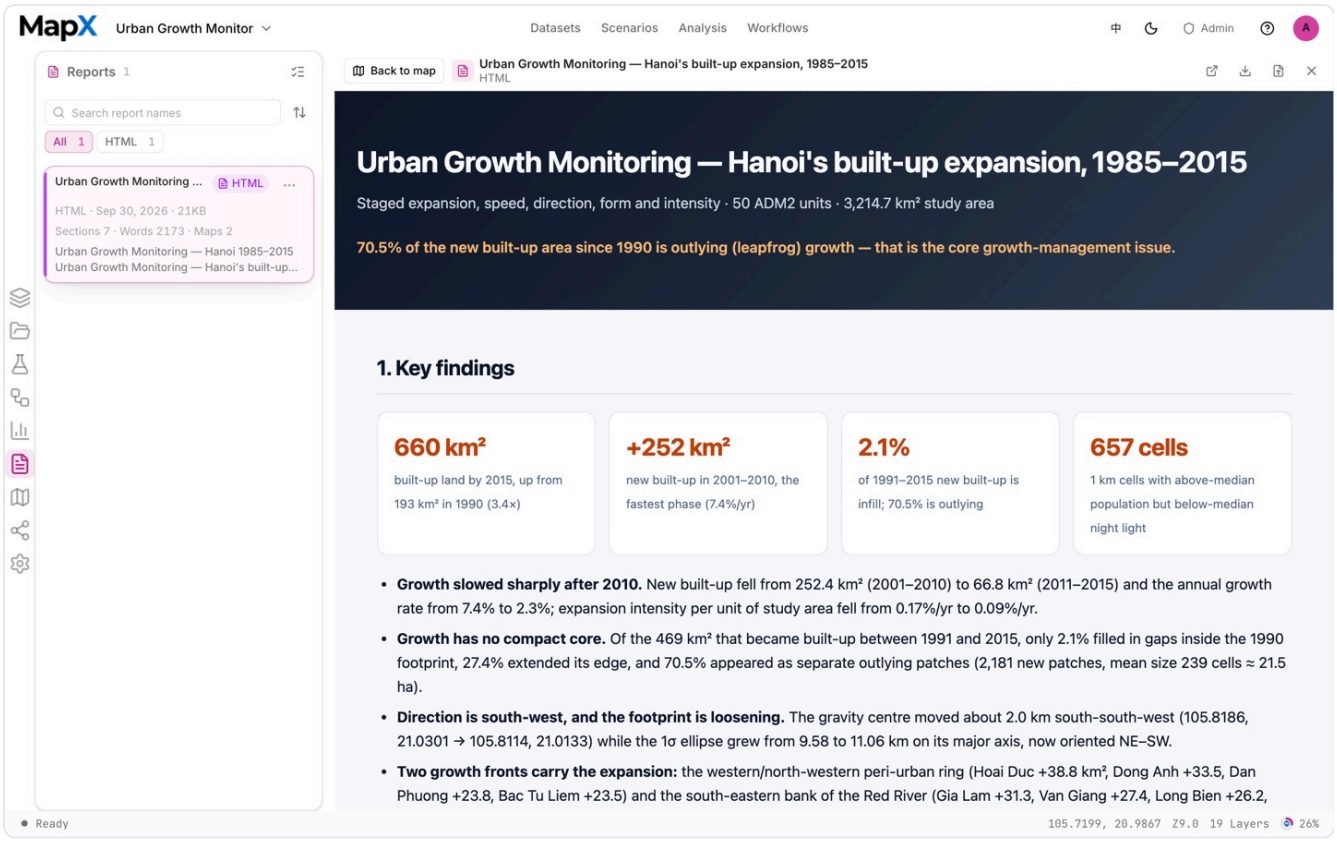
The style editor on a deliverable layer: classed-colour breaks and label settings, with the layer editing panel highlighted in red

5. Turn the result into charts and a report

The run produces a dual-axis growth chart, a form-composition donut and a quadrant distribution. All are **charts** you can restyle or pin, and the **report** records the phase table, the form shares and the year-mismatch caveats.



A chart from this scenario in the chart editor, with the editing area highlighted in red



The scenario report opened in the report canvas, with an executive summary, figures and embedded maps

Example output

> **NOTE**
>
> The screenshots above come from one example run with the default settings on the shipped data. They show the shape of the deliverables, not what your run will return.

A finished run hands back a named, reusable set of deliverables instead of a single map:

- **Layers** — the expansion-form layer (infill, edge, outlying), the population and night-light quadrant grid, the phase rasters as context and a labelled district reference.
- **Tables** — the phase metrics (new area, annual growth, intensity, compactness, centre and ellipse) and the built-year cross table.
- **Charts** — the phase growth chart, the expansion-form donut and the quadrant distribution.
- **Report** — the growth-monitoring document with the phase definitions and the reference-year caveats.

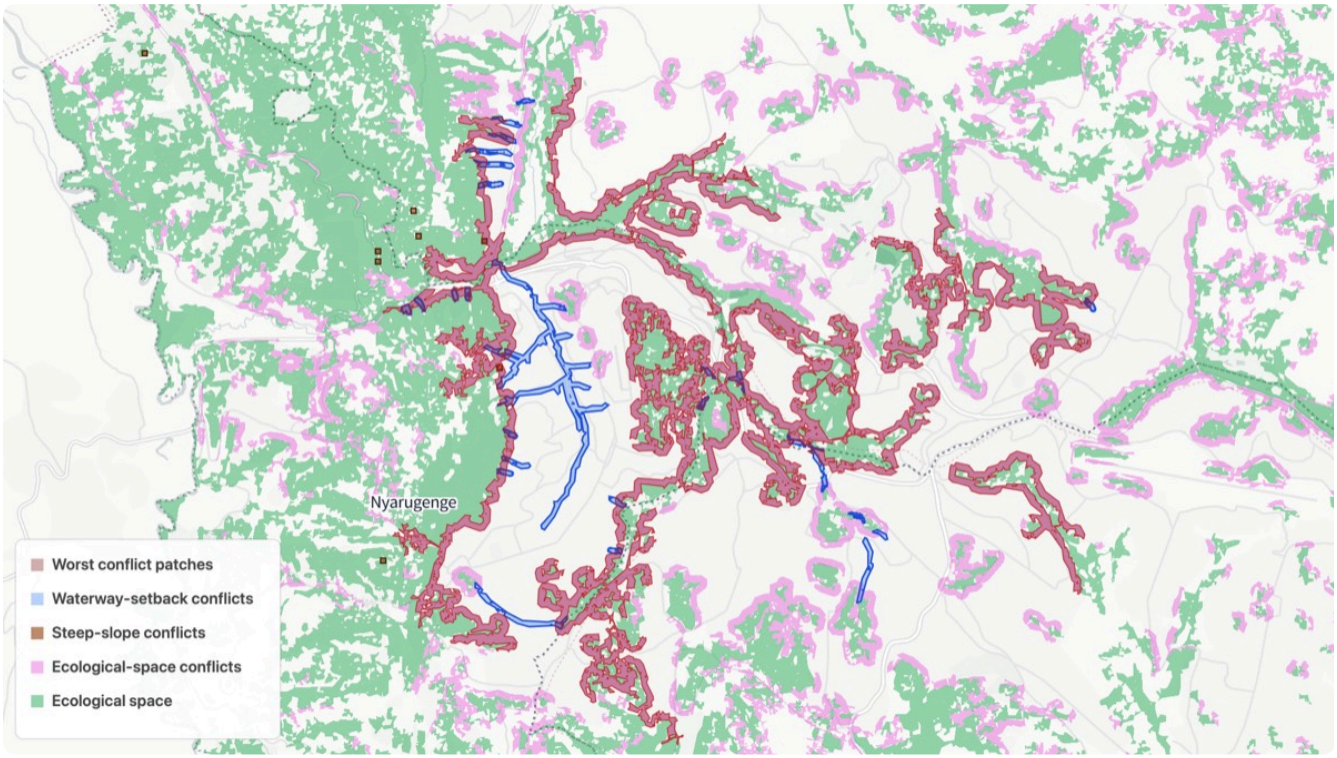
The layer names follow the run language, but the deliverable set is stable enough to plan a handover around.

Open **Urban Growth Monitor** from the scenario catalog at **app.mapxagent.com** and start from its suggested question.

SCENARIO 06

Land-use Conflict Screening Kigali

Find where Kigali's development presses on ecological space, steep slopes and riverbanks, grade the conflicts, and turn them into remediation priorities.



The Kigali run as delivered: ecological space with the three conflict families and the worst-patch ranking

AT A GLANCE

STUDY AREA **Kigali**	ENTRY LAYERS **8 layers**
KEY METHODS **Conflict screening · grading and prioritisation · remediation guidance**	DELIVERABLES **Four conflict layers, a worst-patch ranking, an exposure grid and a report**

Land-use Conflict Screening checks whether a hilly city's built-up land collides with the constraints it should respect: **ecological space, steep slopes, riverbanks and protected areas**. It reports each conflict family on its own and then a de-duplicated union, so one patch cannot be counted twice, and it ends with remediation priorities instead of a single "risk score".

What the scenario ships

Eight layers cover the Kigali window. Only land cover is visible by default; the other layers are analysis inputs and cross-checks.

Layer	Data	Role in the study
Watercourses	OSM waterways, 416 lines	30 m setback corridor
Administrative boundaries	8 ADM3 units	Study area, attribution and labels
Protected area constraints	Overture, 14 features in Rwanda; 1 inside the city	Background and cross-check for ecological space
Building footprints	Overture, 644,665 polygons	Context layer; not used as the development surface (see below)
Land cover	ESA WorldCover 2021, 10 m, 11 classes (visible)	Ecological classes and the built-up surface
Terrain	SRTM 30 m, clipped to the city	Slope classes
Surface water	JRC Global Surface Water, 0–99% frequency	Water presence cross-check
Population	GHSL, 100 m	Exposed population per assessment cell

*All layers come from the **built-in catalog** with source and attribution attached, and they are not limited to Kigali: the catalog provides the same datasets for other cities at the same specification — switch the study area, add the same entries, and this chapter's method chain transfers in minutes.*

How the study works

Stage	Question	What MapX runs	Output
1. Identify the conflicts	Where does built-up land meet each constraint?	Raster calculator, Raster to polygons, Overlay clip, Euclidean distance (100 m / 30 m), Buffer, Slope, Reclassify	Ecological-space mask, steep-slope classes and the four conflict families
2. Grade and prioritise	Which conflicts matter most, and where do they cluster?	Append features, Zonal summary on a 500 m grid, Top N patch ranking, Label anchors	Conflict area per cell, exposed population, worst-patch ranking
3. Remediate	What should be done per type and severity?	Select features, Zonal statistics, Charts, Report	Remediation table, charts and a compliance report

In short, the study first builds spatial masks for each constraint — ecological space, steep slopes and riverbank corridors — and overlays them with the development surface to produce four conflict layers; it then summarises conflict area and affected population on a grid and ranks the worst patches; finally it issues remediation and avoidance guidance by conflict type and grade, delivering conflict layers, a ranking and a screening report.

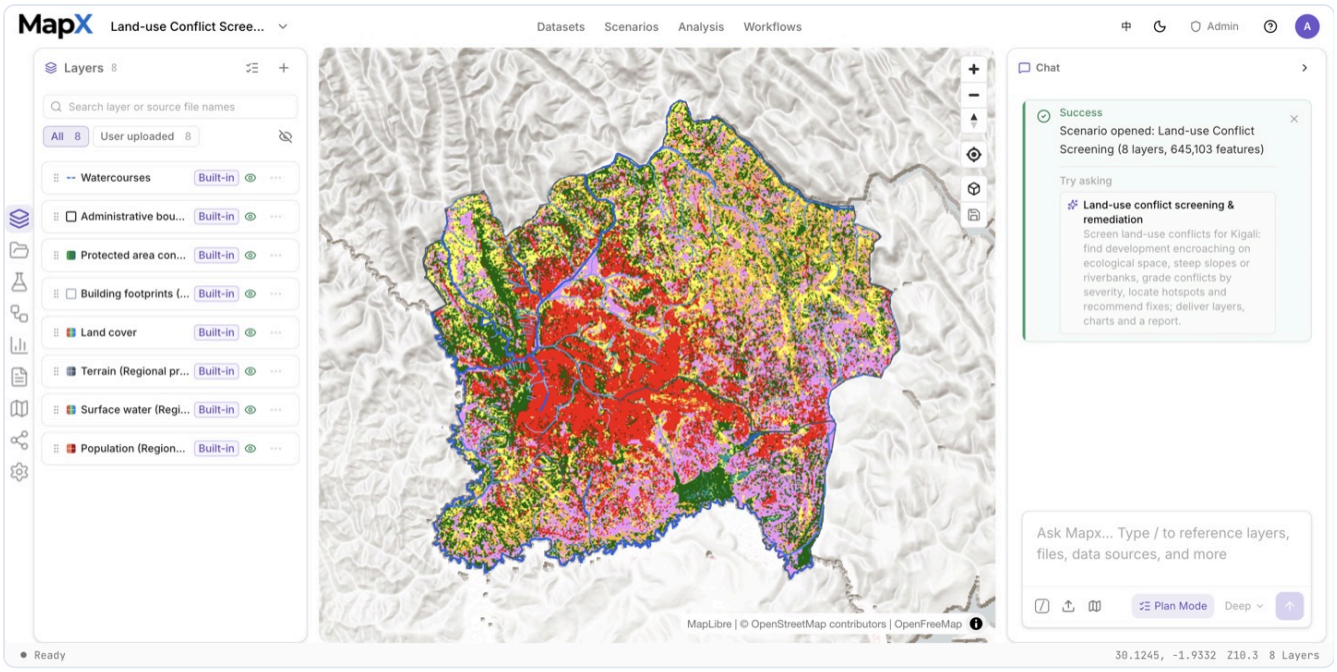
Run it

The screenshots below come from this scenario's reference workspace. Each step names the platform surface it uses, so you can follow the same path in your own project.

> **SUGGESTED QUESTION**
>
> *"Screen land-use conflicts for Kigali: find development encroaching on ecological space, steep slopes or riverbanks, grade conflicts by severity, locate hotspots and recommend fixes; deliver layers, charts and a report."*

1. Open the scenario

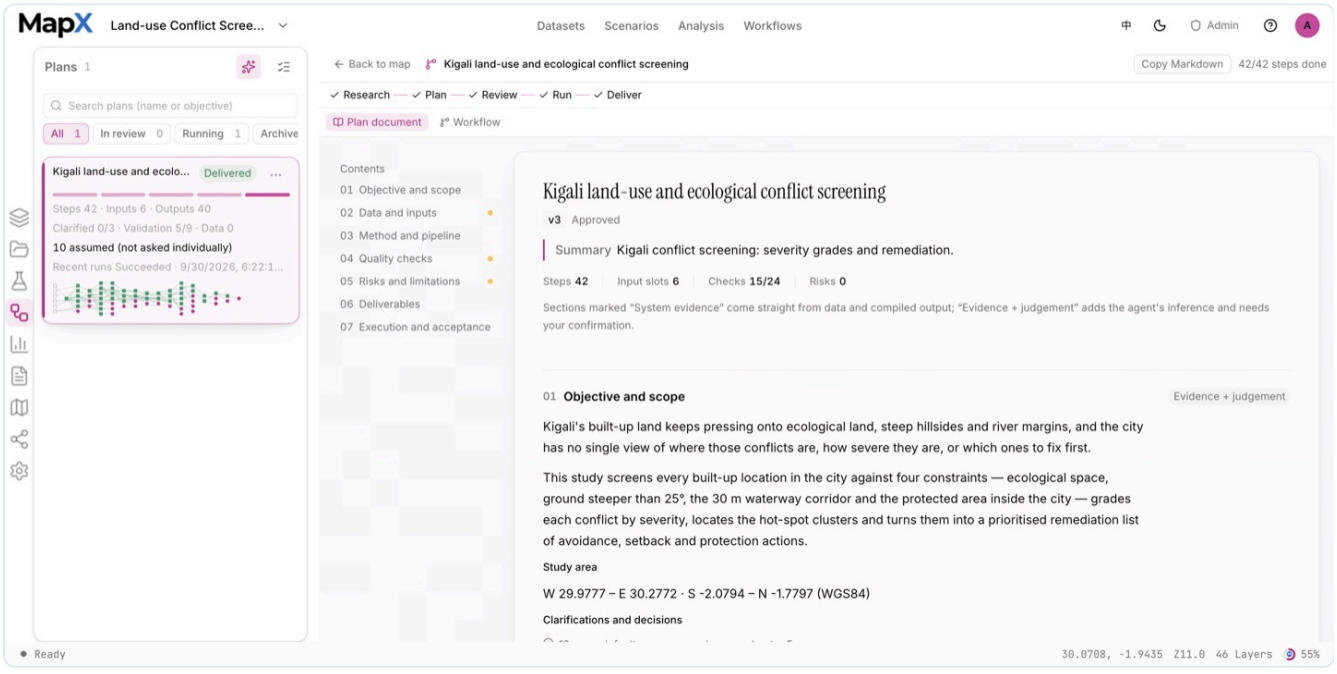
Open **Scenarios** in the top menu, select this card and choose **Open scenario**. The whole bundle imports at once: every layer keeps its default style, legend, source and attribution, and the camera flies to the study area. Use the **Layers** panel to switch layers on and off before you type anything.



The Kigali scenario opened in a fresh project: the layer bundle imported, the Layers panel expanded, and the chat panel confirming the import with the suggested question

2. Ask the suggested question and review the plan

Send the **suggested question** in the box above — or edit it first. **Plan mode** records the ecological-space definition, the development surface and the rule that conflict families are reported per type plus a de-duplicated union.

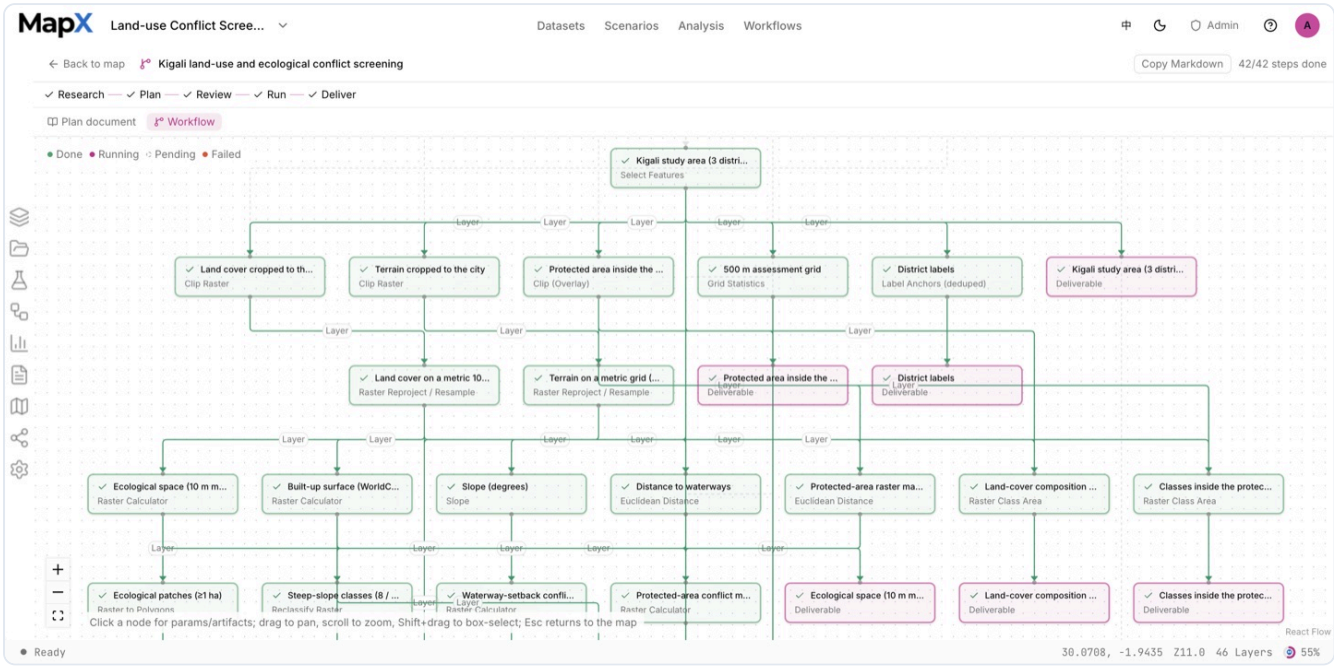


The Kigali plan document with the Plans panel open on the left; plan sections and recorded decisions readable

3. Approve the plan and run the analysis

The chain uses raster algebra and polygon conversion to build the ecological-space layer, Euclidean distance for the 100 m encroachment band and the 30 m waterway setback, buffering, slope reclassification, the

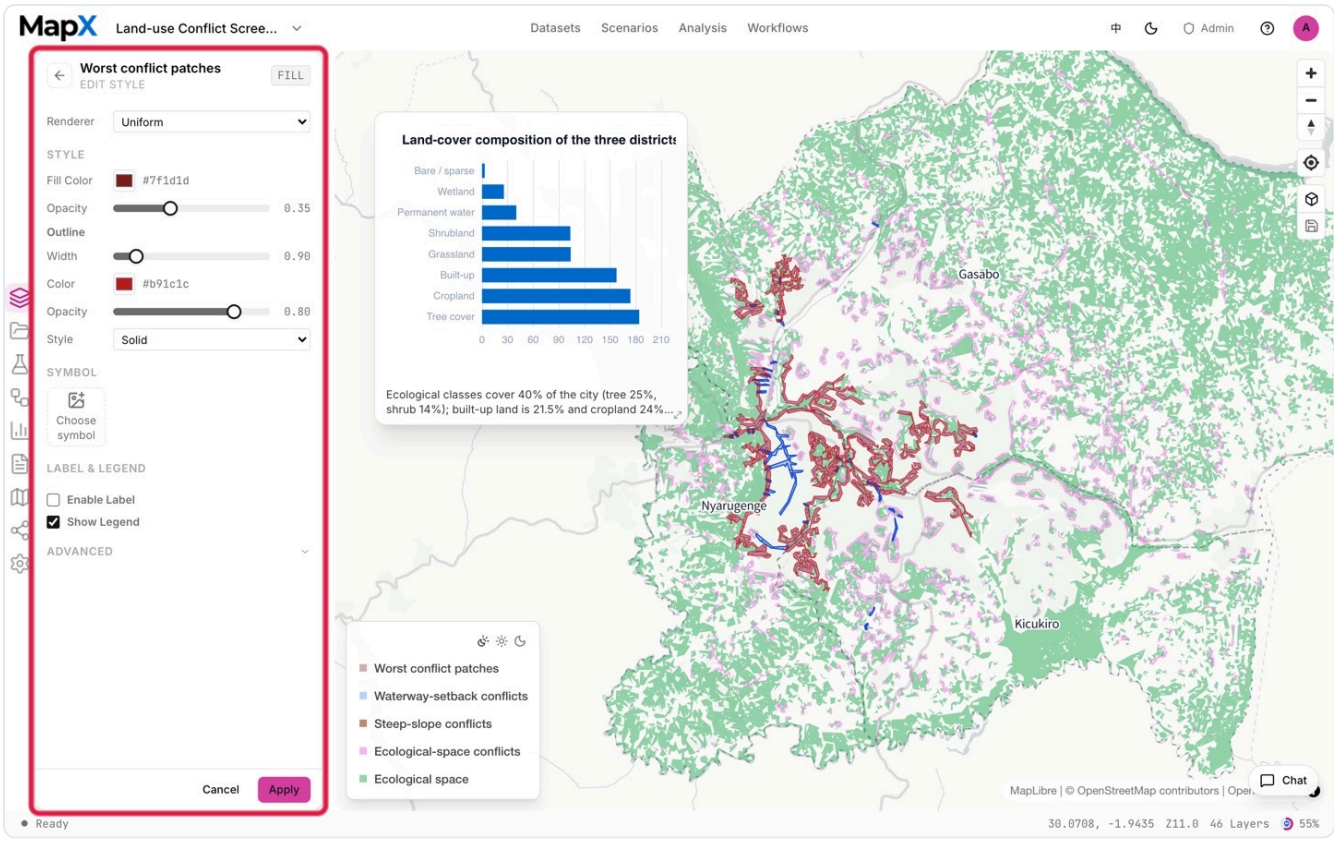
steep-slope mask, and the 500 m grid where conflict area and exposed population are summed. The **Analysis** panel keeps every run's operator, inputs, parameters and result layer, and the workflow graph in the centre visualises the whole chain.



The analysis view for Kigali: the Analysis panel lists every operator run on the left, with the workflow graph in the centre

4. Style the result on the map

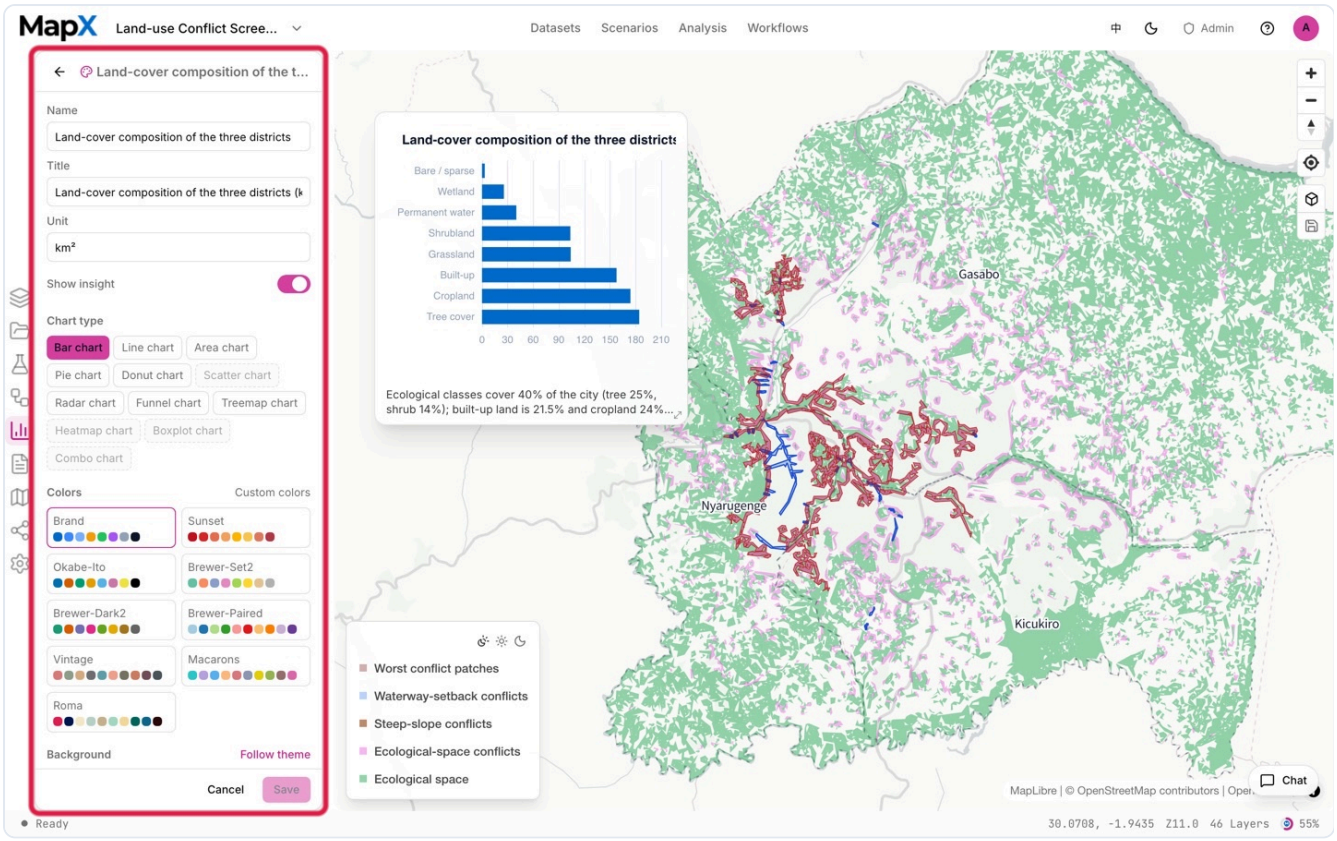
Each conflict family owns one hue so the four translucent layers stay readable together; the worst-patch ranking is styled as a dark-red fill with rank labels from an anchor layer, and legend titles carry the definitions.



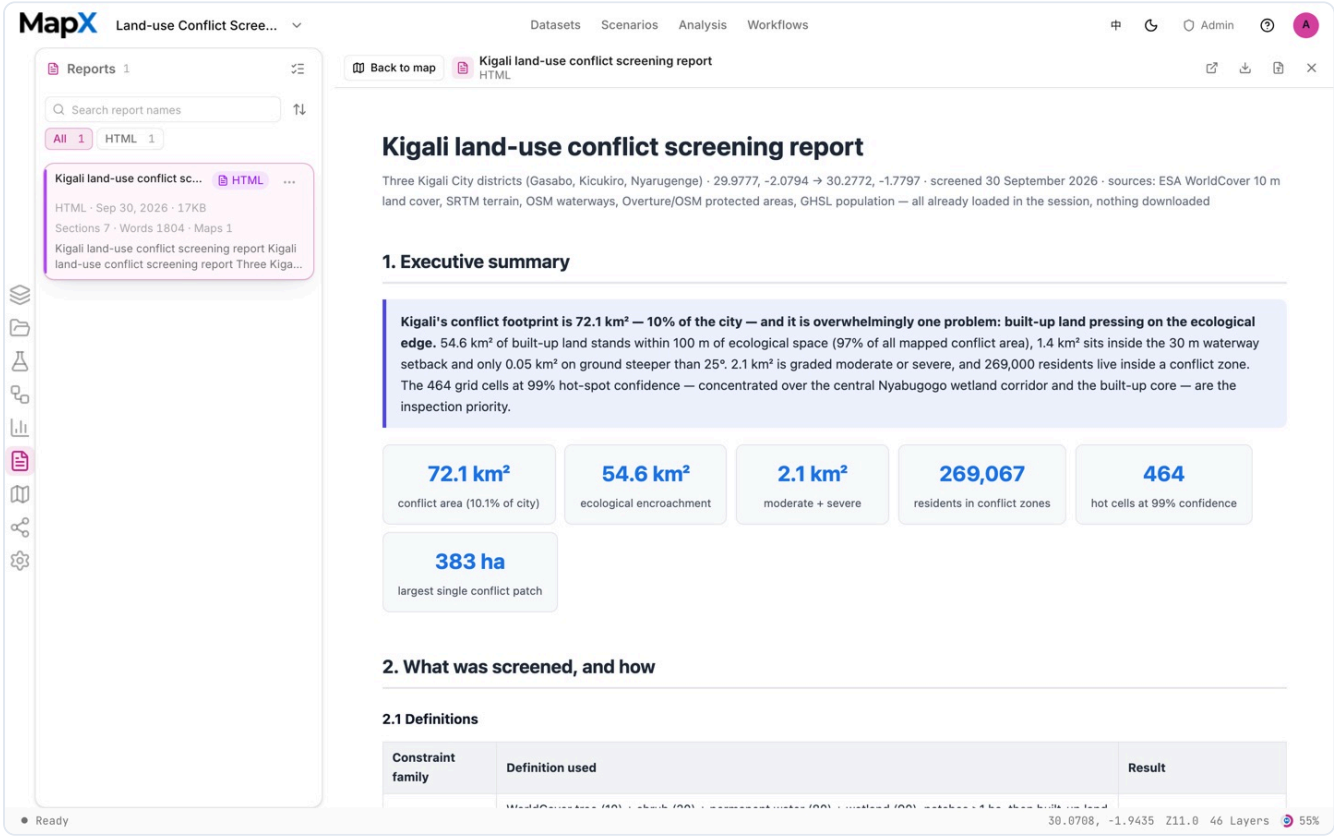
The style editor on a deliverable layer: classed-colour breaks and label settings, with the layer editing panel highlighted in red

5. Turn the result into charts and a report

The run reports conflict area per family and the exposed population, with the union stated separately. The **charts** and the **report** quote those definitions rather than a single risk score.



A chart from this scenario in the chart editor, with the editing area highlighted in red



The scenario report opened in the report canvas, with an executive summary, figures and embedded maps

Example output

> **NOTE**
>
> The screenshots above come from one example run with the default settings on the shipped data. They show the shape of the deliverables, not what your run will return.

A finished run hands back a named, reusable set of deliverables instead of a single map:

- **Layers** — the ecological-space layer, the four conflict families (ecological encroachment, steep slope, waterway setback, protected area) and the worst-patch ranking, all with their definitions in the legend.
- **Tables** — the per-type conflict areas plus the de-duplicated union, and the exposed population per assessment cell.
- **Charts** — the conflict-area and exposure summaries, editable and ready for the report.
- **Report** — the compliance document with the definitions, the areas and the remediation guidance.

The layer names follow the run language, but the deliverable set is stable enough to plan a handover around.

Open **Land-use Conflict Screening** from the scenario catalog at **app.mapxagent.com** and start from its suggested question.

APPENDIX A

Appendix A: Scenarios at a glance

Scenario	Study area	Layers	Key methods	Deliverables
1 15-minute Neighbourhood Coverage	Jakarta	7	Walking service areas, shortfall population, 2SFCA	Three shortfall layers, a neighbourhood table, rankings and a heatmap
2 Development Suitability Baseline	Nairobi	8	Hard-constraint screening, weighted overlay, capacity estimate	Suitability classes, development control tiers, charts and a report
3 Carrying Capacity Assessment	Jakarta	6	Fuzzy membership, limiting-factor composite, pressure quadrants	Capacity classes, overload cells, charts and a report
4 Public Facility Siting	Dar es Salaam	7	Coverage diagnosis, candidate screening, MCLP vs k-median	Recommended sites, a marginal coverage curve, a ward balance table
5 Urban Growth Monitor	Hanoi	7	Phased expansion metrics, landscape expansion index, centre and ellipse	Expansion-form layers, phase tables, a quadrant grid
6 Land-use Conflict Screening	Kigali	8	Conflict screening, grading, remediation guidance	Four conflict layers, a worst-patch ranking, an exposure grid

Appendix B: Get-started checklist

- Open the **Scenarios** catalog and pick the card that matches your planning question.
- Choose **Open scenario**: the layer bundle imports with default styles, no upload or configuration.
- Send the scenario's **suggested question** (copy it from the box at the start of each chapter's Run it section).
- Confirm the conventions and approve the plan in **plan mode**, then let it run.
- Export a **report**, refine the **charts** or share the map, and keep asking follow-ups to adjust scope and thresholds.

Appendix C: Data and licences

Scenarios are assembled from the built-in data catalog, and every layer carries its source, licence and attribution. Land cover, population, built-up area, protected areas and road networks each have their own vintage and resolution; state the year and scale whenever you combine them. Scenario output is analysis, not a statutory planning document.

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